PEARSON EDEXCEL INTERNATIONAL A LEVEL PHYSICS · WPH11

Unit 1: Mechanics and Materials

Revision notes, spec point by spec point (Mechanics and Materials, points 1–32)

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

How to pick up the marks.

MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

What examiners keep seeing go wrong.

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Topic: Mechanics

SPEC 1

Equations of uniformly accelerated motion

WHAT THIS SPEC POINT ASKS

Use the four equations of motion (suvat) for motion in a straight line with constant acceleration.

1 The five quantities

Symbol	Quantity	Unit	Vector or scalar?
s	displacement	m	vector
u	initial velocity	m s^{-1}	vector
v	final velocity	m s^{-1}	vector
a	acceleration	m s^{-2}	vector
t	time	s	scalar

1 The four equations (all in the data booklet)

$$v = u + at \qquad (\text{no } s)$$

$$s = ut + \tfrac{1}{2}at^2 \qquad (\text{no } v)$$

$$v^2 = u^2 + 2as \qquad (\text{no } t)$$

$$s = (u + v)t \div 2 \qquad (\text{no } a)$$

- They work **only when acceleration is constant** (this includes a = 0). Clues in a question: "constant force", "uniform acceleration", "falls freely", "air resistance is negligible".
- They **don't** work when the acceleration changes: a curved slide, a car with a varying driving force, anything with significant drag. Use energy or a graph instead.
- Where they come from: on a v–t graph with constant acceleration, the gradient is (v − u)/t = a, which gives v = u + at. The area (a trapezium) is s = (u + v)t/2. Combine the two and you get the other equations.

1 Method

1. Sketch the situation. **Choose a positive direction** and stick to it (e.g. up = +, so g = −9.81 m s^{-2}).
2. List s, u, v, a, t. Write the three you know and the one you want.
3. Pick the equation that **leaves out** the quantity you neither know nor want.
4. Rearrange, then substitute with units. Give the answer to the same s.f. as the data (2 or 3).

Hidden values: "from rest" or "dropped" means u = 0. "Comes to rest" means v = 0. "Free fall" means a = g = 9.81 m s^{-2} downwards. At the **top** of a vertical throw, v = 0.

WORKED EXAMPLE: BRAKING DISTANCE

A car travelling at 25 m s^{-1} brakes with constant deceleration and stops in 40 m. Find the deceleration and the time taken.
Known: u = 25 m s^{-1}, v = 0, s = 40 m. Want a (no t needed) → $v^2 = u^2 + 2as$.
$0 = 25^2 + 2 \times a \times 40$ → a = −625 ÷ 80 = **−7.8 m s^{-2}** (the minus sign means it is slowing down).
Time: v = u + at → t = (0 − 25) ÷ (−7.81) = **3.2 s**.

WORKED EXAMPLE: A QUADRATIC IN T

A stone is thrown vertically upwards at 8.0 m s^{-1} from the edge of a cliff 20 m above the sea. How long until it hits the sea?
Take **up as positive**: u = +8.0 m s^{-1}, a = −9.81 m s^{-2}, s = −20 m (it ends up below the start).
$s = ut + \frac{1}{2}at^2$ → $-20 = 8.0t - 4.905t^2$ → $4.905t^2 - 8.0t - 20 = 0$.
$t = [8.0 \pm \sqrt{(8.0^2 + 4 \times 4.905 \times 20)}] \div (2 \times 4.905)$ → t = 2.99 s or −1.36 s.
A negative time is not physical, so **t = 3.0 s**.

HOW WPH11 MARKS CALCULATIONS (APPLIES TO THE WHOLE UNIT)

- **Units**: there is no separate unit mark, but a missing or wrong unit normally loses the **final** mark. It is charged once per question part. "Show that" answers are exempt when the unit is given.
- **g**: use 9.81 m s^{-2} (9.8 is fine). Using **g = 10 loses a mark**.
- **Significant figures**: too many s.f. is never penalised if it rounds to the right value. In a **"show that"**, give **one more s.f.** than the printed value (show that ≈ 3.4 → write 3.42). If a question says "to at least 3 s.f.", that is a mark.
- **Rounding**: round, never truncate, and carry full calculator values forward. Rounding 0.1386 s to 0.138 s lost a mark (WPH11 Jan 2024 Q12). Final answers as decimals with units, not fractions or surds.
- **"Show that"**: write the equation, the substituted numbers and the answer. If you can't get the printed value, use it anyway in the next part: those marks are still available.
- **"Deduce / determine whether / show whether"**: calculate, then **compare both numbers (with units)** and state a conclusion in the question's words. This is the most-lost mark in the unit, according to almost every examiner report.
- "Use of $v^2 = u^2 + 2as$" style method marks are given when the equation is seen or implied by correct substitution, even with a power-of-ten slip. Define any symbol the question hasn't.
- If a published mark scheme seems to disagree with the numbers printed in the question, trust the question: a few schemes contain typing errors.

MARK-SCHEME LANGUAGE

- "Use of $v^2 = u^2 + 2as$" (or any suvat equation) is the method mark. "Allow use of any valid suvat method."
- Sign convention counts: swapping u and v to get a positive deceleration loses the mark (WPH11 Jan 2019 Q16). "u and a having the same sign convention" (WPH11 Jan 2026 Q18).
- "u = 0" must be **stated** when a printed answer relies on it.

Often asked as: multi-step calculations (probe slowing down, fall time on Mars), "show that" a launch speed, comparing two times in a race, "why can't the equations of motion be used here?" (acceleration not constant), MCQ rearrangements such as $a = (v^2 - u^2)/2s$.

COMMON MISTAKE

Using the wrong v in a multi-stage story (e.g. using v = 0 at the end when the question is about the speed at the end of the first stage).
Using a = g for **horizontal** motion. Horizontally, a = 0 unless something pushes.
Doubling the time to the top to get the total flight time when the object lands **below** (or above) its launch height.
Mixing signs: if up is positive, a = −9.81 m s^{-2} for the whole flight, up **and** down.

BEYOND THE SPEC, BUT EXAMINED

- Solving the **quadratic** from $s = ut + \frac{1}{2}at^2$ and choosing the physically sensible root (WPH11 Jun 2022 Q18).
- Turning a rate into a time: drips falling at 5 per second are 0.2 s apart, so four gaps = 0.8 s (WPH11 Jun 2019 Q18).
- Strobe and video frames: the time between images is 1 ÷ (flash or frame rate) (WPH11 Oct 2024 Q14).

SPEC 2

Displacement–time, velocity–time and acceleration–time graphs

WHAT THIS SPEC POINT ASKS

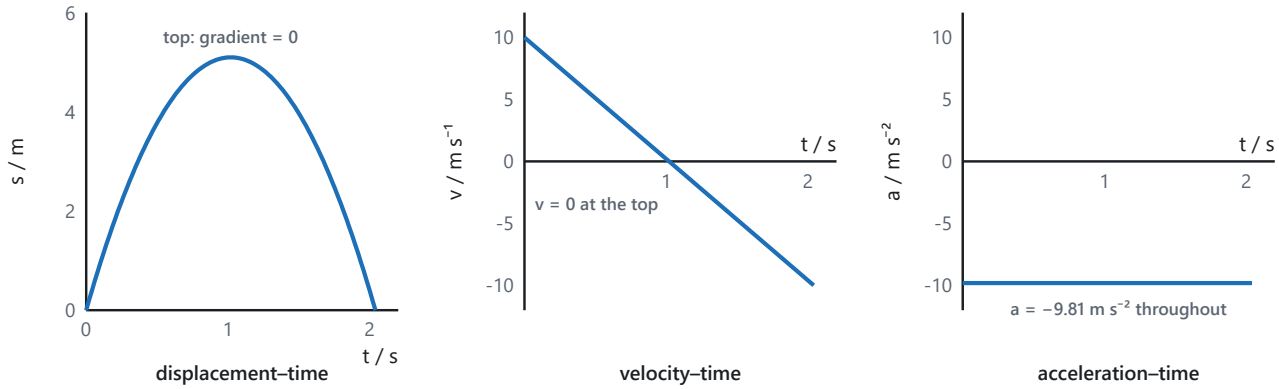
Draw and interpret s–t, v–t and a–t graphs.

2 Reading each graph

Graph	Horizontal line means	Straight sloping line	Curve
s–t	at rest (gradient = 0, so v = 0)	constant velocity	changing velocity (accelerating)
v–t	constant velocity (a = 0)	constant acceleration	changing acceleration
a–t	constant acceleration	acceleration changing at a steady rate	

- The **sign** matters. On an s–t graph a negative gradient means moving **back towards the start**, not slowing down. On a v–t graph, a line crossing the time axis means the object **stops and reverses**.
- "Decelerating" only means the speed is falling. An object whose velocity goes from +5 to 0 to −5 m s^{-1} is slowing down, stopping, then speeding up in the opposite direction.
- To get one graph from another: the v–t graph is the **gradient** of the s–t graph at each moment; the a–t graph is the gradient of the v–t graph.

2 A ball thrown straight up and caught (up = positive)



Ball thrown up at 10 m s^{-1}. s–t: a parabola, gradient zero at the top. v–t: a straight line of gradient −9.81 m s^{-2}, crossing zero at the top. a–t: constant at −9.81 m s^{-2}, including at the top.

- The velocity **changes sign** at the top; the acceleration never does. At the top, v = 0 but a = −9.81 m s^{-2}.
- For a **bouncing ball** (down = negative), the v–t graph is a set of parallel sloping lines with a near-vertical jump from negative to positive at each bounce.

2 Drawing graphs from data

- Axes labelled with **quantity / unit**, the right way round (the variable you control on x).
- A scale that uses **more than half** of each axis, in steps of 1, 2 or 5 (never 3 or 7).
- Plot to within half a small square; draw a thin **line of best fit** (a smooth curve if the trend curves).
- On a **sketch**, put the key values from earlier parts on the axes and show asymptotes (e.g. terminal velocity).

MARK-SCHEME LANGUAGE

- "Gradient increasing between 0 and T_A; straight line with positive gradient between T_A and T_B" (WPH11 Oct 2025 Q18).
- "Curved line starting at zero with gradient decreasing; horizontal line after …" for an object reaching terminal velocity (WPH11 Jun 2025 Q20).
- A flat s–t line: say "**gradient = 0, so velocity = 0**", not just "the line is flat".
- When asked to describe the motion from a graph, every point must **refer to the graph** (gradient, area, value); an answer that never mentions the graph scores at most 1.

Often asked as: describe the motion shown; convert an s–t graph into a v–t graph with a scale; sketch v–t for a falling object; MCQs on which graph matches a story.

COMMON MISTAKE

Saying an object is "decelerating from 9 s onwards" when it actually stops and **reverses** (WPH11 Jan 2019 Q11).
Reading a negative s–t gradient as "slowing down".
Drawing a straight line where the motion needs a curve, or a curve bending the wrong way.
Plotting a scale that fills less than half the grid: that is a lost mark.

BEYOND THE SPEC, BUT EXAMINED

- Axis-scaling is a mark on its own: the scale must fill at least half of each axis (WPH11 Jun 2019 Q13).
- Sketching an unfamiliar relationship (e.g. launch speed against angle) as a curve with an asymptote, with a value from an earlier part marked on it (WPH11 Jun 2019 Q14).

SPEC 3 Gradients and areas of motion graphs, including non-uniform acceleration

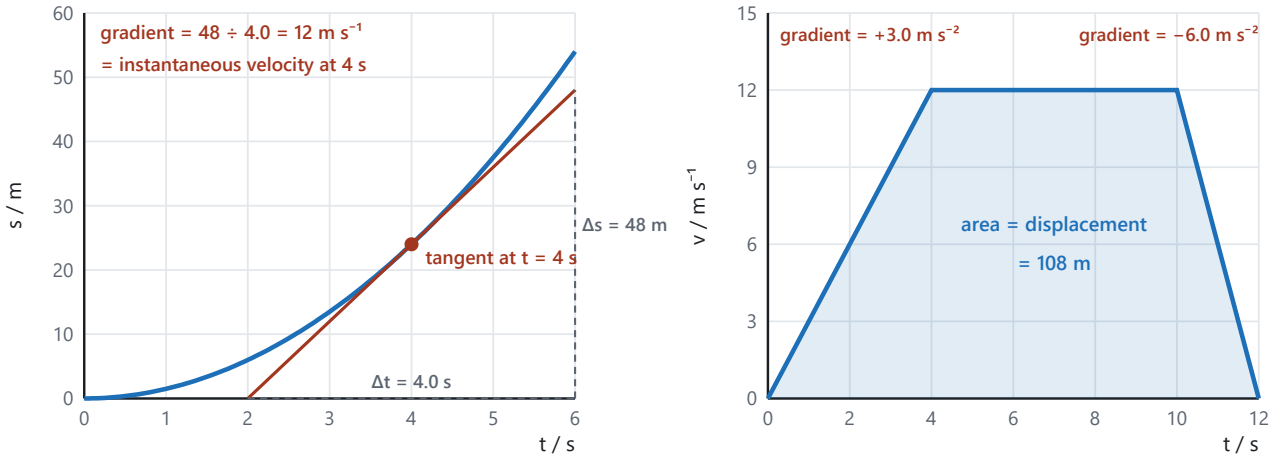
WHAT THIS SPEC POINT ASKS

Know what the gradient and the area of each motion graph represent, for constant and changing acceleration, and use them.

3 What the gradient and the area give

Graph	Gradient =	Area under the line =
s–t	velocity (m ÷ s = m s^{-1})	(nothing useful)
v–t	acceleration (m s^{-1} ÷ s = m s^{-2})	**displacement** (m s^{-1} × s = m)
a–t	(rate of change of a, not needed)	**change in velocity** (m s^{-2} × s = m s^{-1})

A quick check: multiply the units of the two axes to get the unit of the area; divide them to get the unit of the gradient.



Left: on a curved s–t graph, the gradient of the tangent gives the velocity at that instant. Right: the gradient of each section of the v–t graph is the acceleration; the shaded area is the displacement.

3 Instantaneous and average rates

- **Average** velocity = total displacement ÷ total time. On a graph this is the gradient of the straight line (a **chord**) joining two points.
- **Instantaneous** velocity (the value at one moment) on a curved s–t graph = the gradient of the **tangent** at that point.
- Drawing a tangent: use a ruler; the line should touch the curve at the point and make equal angles with the curve on either side. Then make a **large triangle** (more than half the length of the drawn line) and read Δy and Δx from it.
- Same idea on a curved v–t graph: the tangent's gradient is the acceleration at that instant. The **maximum** acceleration is at the steepest point.

WORKED EXAMPLE: VELOCITY FROM A TANGENT (LEFT GRAPH)

The tangent drawn at t = 4.0 s passes through (2.0 s, 0 m) and (6.0 s, 48 m).
gradient = Δs ÷ Δt = (48 − 0) ÷ (6.0 − 2.0) = **12 m s^{-1}**: the velocity at 4.0 s.
The **average** velocity over the first 4.0 s is different: 24 m ÷ 4.0 s = 6.0 m s^{-1} (gradient of the chord from the origin).

WORKED EXAMPLE: AREA UNDER A V–T GRAPH (RIGHT GRAPH)

A cyclist accelerates from rest to 12 m s^{-1} in 4.0 s, rides at 12 m s^{-1} for 6.0 s, then stops in 2.0 s.
Accelerations: 12 ÷ 4.0 = **+3.0 m s^{-2}**; then 0; then (0 − 12) ÷ 2.0 = **−6.0 m s^{-2}**.
Displacement = area = ½ × 4.0 × 12 + 6.0 × 12 + ½ × 2.0 × 12 = 24 + 72 + 12 = **108 m** (1.1 × 10^2 m to 2 s.f.).
Average velocity = 108 ÷ 12 = **9.0 m s^{-1}**.

3 Non-uniform acceleration: areas under curves

- **Count squares**: count whole squares under the curve, add the part-squares (estimate halves and more-than-halves). Then work out what **one square** is worth from the axis scales and multiply.
  - Example: 1 small square = 1 m s^{-1} × 0.5 s = 0.5 m. 86 squares under the curve → displacement ≈ 86 × 0.5 = 43 m.
- **Trapezia**: split the area into vertical strips and treat each as a trapezium (½ × (y_1 + y_2) × width). Good enough when the curve is gentle.
- Area **below** the time axis counts as **negative** displacement. A diver who goes down and comes back to the surface has equal areas above and below the axis: total displacement 0, but the distance travelled is the sum of both areas.
- On an a–t graph, the area is the **change** in velocity. Add the initial velocity to get the final velocity.

MARK-SCHEME LANGUAGE

- Tangent questions: "draws tangent at …", "determines values for Δv and Δt (from the tangent)", "acceleration = …" (three marks, WPH11 Oct 2024 Q19). Dividing a single v reading by t "scored no marks".
- Readings: "no tolerance if readings taken at the stated times; allow ± half a small square otherwise" (WPH11 Jun 2025 Q12).
- Areas: a range is accepted (e.g. "13.5 m to 14.5 m", WPH11 Jun 2026 Q15); state the value of one square.

Often asked as: acceleration at an instant from a tangent; maximum acceleration (tangent at the steepest point); average acceleration between two times; distance from a v–t area with trapezia or squares; an MCQ on areas below the axis.

COMMON MISTAKE

Using total velocity ÷ total time (or one point) when the question needs a gradient: maximum of one mark.
Drawing the tangent at the wrong time, or using a small triangle.
Missing a prefix on an axis (time in **ms**, distance in **km**) (WPH11 Jan 2024 Q16).
Treating the area under an a–t graph as the final velocity, not the change in velocity (WPH11 Jan 2025 Q5).

EXAM TIP

State your working on the graph itself: draw the tangent or triangle, and write the two readings you used. Examiners can then give method marks even if a reading is slightly off.

SPEC 4

Scalars and vectors

WHAT THIS SPEC POINT ASKS

Know the difference between scalars and vectors, give examples of each, and recognise vector notation.

4 Definitions

- A **scalar** has magnitude (size) only.
- A **vector** has magnitude **and direction**.

Scalars	Vectors
distance, speed, time, mass, energy (all types), work, power, density, temperature, volume, efficiency	displacement, velocity, acceleration, **force** (including **weight**, friction, tension, upthrust, drag), momentum, gravitational field strength

- **Distance** is the length of the path travelled. **Displacement** is the straight-line distance from start to finish, **in a stated direction**.
- **Speed** = distance ÷ time. **Velocity** = displacement ÷ time (a speed in a direction). A car going round a bend at constant speed has a **changing velocity**, so it is accelerating.
- In one dimension, the direction is shown by the **sign**: +3 m s^{-1} and −3 m s^{-1} have the same speed but opposite directions.

4 Vector notation

- In print a vector is written in **bold** (often italic), e.g. ***F***, or with an arrow over the letter (F⃗). By hand: an arrow above or a line under the symbol.
- A vector equation like **W** + $\mathbf{T_1}$ + $\mathbf{T_2}$ = 0 means the forces, **added as vectors** (head to tail), close up into a triangle: the object is in equilibrium. It does **not** mean the sizes add up to zero.

WORKED EXAMPLE: SPEED AGAINST VELOCITY

A runner completes one lap of a 400 m track in 50 s.
Average speed = 400 ÷ 50 = **8.0 m s^{-1}**. Displacement = 0 (back at the start), so average velocity = **0**.

MARK-SCHEME LANGUAGE

- Distance v displacement (2 marks): "displacement is the (straight-line) distance **in a (specified) direction**"; "distance is the length of the path travelled". Fallback for 1 mark: "displacement is a vector, distance is a scalar" (WPH11 Jun 2025 Q12).

Often asked as: MCQ "which of these is a vector / scalar?" (almost every paper); distance v displacement; reading a vector equation for forces in equilibrium.

COMMON MISTAKE

The classic MCQ trap: **weight is a vector** (it is a force), **mass is a scalar**. Momentum and acceleration are vectors. Energy, work, power and efficiency are scalars.
Reading "kWh" as "kilowatts per hour": it is a unit of **energy** (kW × h = 1000 W × 3600 s = 3.6×10^6 J) (WPH11 Jan 2019 Q2).

BEYOND THE SPEC, BUT EXAMINED

- Unit equivalences: kg m s^{-2} = N; N s = kg m s^{-1} (momentum); N kg^{-1} = m s^{-2} (WPH11 Jan 2022 Q10).

SPEC 5

Resolving a vector into components

WHAT THIS SPEC POINT ASKS

Split a vector into two perpendicular components, by scale drawing and by calculation.

5 The rule

A vector F at angle θ to the horizontal has the same effect as two perpendicular vectors acting together:

$$\text{horizontal component} = F\cos\theta \qquad \text{vertical component} = F\sin\theta$$

- **cos goes with the component next to (adjacent to) the angle.** If the angle is measured from the **vertical**, the vertical component is F cos θ and the horizontal one is F sin θ.
- Always sketch the right-angled triangle first and mark which angle you've been given.
- **By drawing**: choose a scale, draw the vector at the right angle with a protractor, drop perpendiculars onto the horizontal and vertical lines, measure each component and convert with the scale.

5 Weight on a slope

For an object of weight mg on a slope at angle θ to the horizontal:

$$\text{component down the slope} = mg\sin\theta \qquad \text{component into the slope} = mg\cos\theta$$

Sanity check: when θ → 0 (flat), the down-slope part → 0. When θ → 90°, all the weight is down the "slope".

WORKED EXAMPLE: RESOLVING A FORCE

A rope pulls a sledge with a force of 250 N at 35° above the horizontal.
Horizontal = 250 cos 35° = **205 N** (forwards). Vertical = 250 sin 35° = **143 N** (upwards).

WORKED EXAMPLE: ON A SLOPE

A 60 kg skier stands on a slope at 20° to the horizontal. Weight = 60 × 9.81 = 589 N.
Down the slope: 589 sin 20° = **201 N**. Into the slope: 589 cos 20° = **553 N** (the normal contact force balances this).

MARK-SCHEME LANGUAGE

- "Use of appropriate trigonometry" is a mark of its own (often with ecf); the calculation mark then needs the correct function.
- A **percentage gradient** is a tangent: a 20 % slope rises 20 m per 100 m horizontally, so $\theta = \tan^{-1}(0.20) = 11.3°$ (WPH11 Jan 2019 Q18).
- Two ropes or two halves of a wire sharing a load: $2T \sin\theta = W$ (or $2T \cos\theta$, depending on the angle given) (WPH11 Jan 2024 Q6, Jan 2025 Q10).

Often asked as: launch-velocity components (often "show that" or "to at least 3 s.f."); weight down a slope; forward component of an angled pull; tension in two supporting ropes.

COMMON MISTAKE

sin and cos swapped; calculator in **radians** (always check DEG).
Giving the complementary angle (62° instead of 28°).
Forgetting the factor of 2 when two ropes share the load.

SPEC 6

Adding two vectors: the resultant

WHAT THIS SPEC POINT ASKS

Find the resultant of two coplanar vectors at any angle by scale drawing, and at right angles by calculation.

6 The resultant

The **resultant** is the single vector with the same effect as all the vectors together.

6 At right angles: calculate

$$R = \sqrt{(A^2 + B^2)} \qquad \tan\theta = B \div A$$

Give the **direction** too, and say what the angle is measured from ("67° from the 5.0 N force", "north of east").

WORKED EXAMPLE: RESULTANT AT RIGHT ANGLES

Forces of 5.0 N east and 12 N north act on a point.
$R = \sqrt{(5.0^2 + 12^2)} =$ **13 N**. $\theta = \tan^{-1}(12 \div 5.0) =$ **67° north of east**.

6 At any other angle: scale drawing (triangle method)

1. Choose a scale that makes the diagram big (e.g. 1 cm = 10 N, so the longest line is at least 6–8 cm). **Write the scale down.**
2. Draw the first vector to scale with a ruler, in the right direction, arrow in the middle of the line.
3. From its **head**, draw the second vector (tip to tail) at the correct angle, using a protractor.
4. The resultant goes from the **tail of the first** to the **head of the second**. Measure its length, convert with the scale, and measure its angle.

Parallelogram method: draw both vectors from the same point, complete the parallelogram, and the diagonal is the resultant. To **subtract** a vector, add its reverse.

MARK-SCHEME LANGUAGE

Typical 4 marks for a scale drawing:

- "Straight line at least 6 cm long representing … with label" (scale stated).
- "Vector triangle with at least two sides labelled"; "all three arrows in the correct relative directions".
- "Magnitude of resultant = … (allow a range, e.g. 370–390 N)".
- "Correct answer from trigonometry scores the last mark only." A calculated answer where a **drawing** was asked gets at most one mark.

Often asked as: resultant of two forces (e.g. two tugs pulling a ship); scaled momentum diagrams; Pythagoras MCQs; "explain how you would measure the angle".

COMMON MISTAKE

Drawing a free-body diagram (arrows from a point) instead of a **tip-to-tail triangle**.
Labelling a side with its drawn length ("8 cm") instead of the quantity ("70 N").
Arrowheads at the ends of lines overlapping the next vector: put them in the middle.
The resultant arrow pointing the wrong way. Blunt pencil, no ruler or protractor.

SPEC 7

Projectiles: independent horizontal and vertical motion

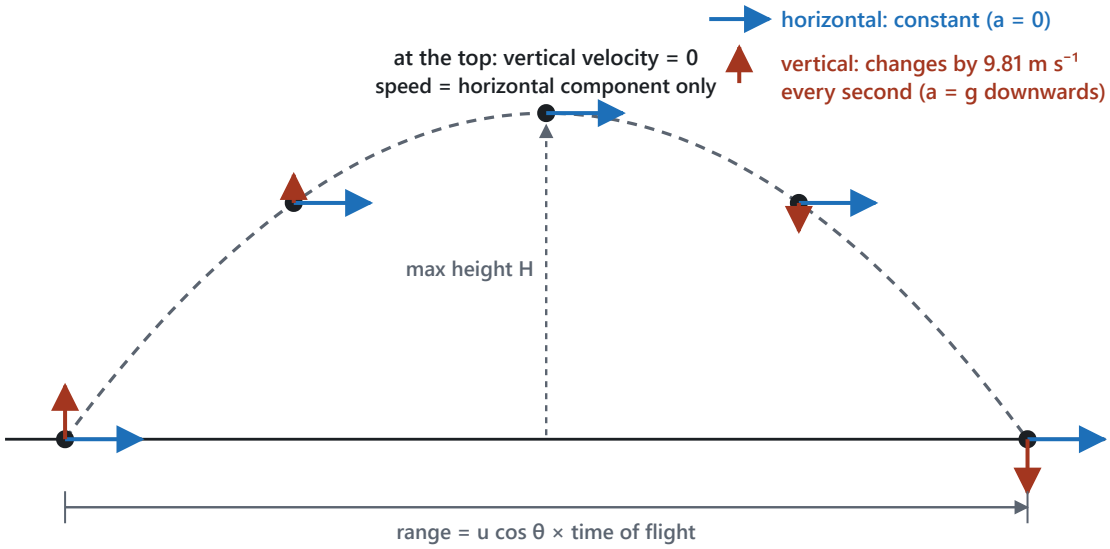
WHAT THIS SPEC POINT ASKS

Use the fact that the horizontal and vertical motions of a projectile moving freely under gravity are independent of each other.

7 The model (air resistance negligible)

	Horizontal	Vertical
Force	none	weight only
Acceleration	**0**	**g = 9.81 m s^{-2} downwards**
Velocity	**constant** = u cos θ	changes by 9.81 m s^{-1} every second
Equation	s = ut	any suvat equation with a = −9.81 (up +)

The **time** is the same for both. That's the link: find t from one direction and use it in the other.



The horizontal arrow stays the same length all the way. The vertical arrow shrinks to zero at the top, then grows downwards.

Key facts:

- At the **top**, the vertical velocity is 0 but the acceleration is still g downwards. The speed there is just the horizontal component.
- Launched horizontally from a height: $u_{vertical} = 0$, so the fall time is set by the height alone ($s = \frac{1}{2}gt^2$). A ball dropped and a ball fired horizontally from the same height land at the same time.
- Landing at the launch height: the path is symmetrical, so total time = 2 × time to the top. **Only then.**
- The **final velocity** is a vector: combine the horizontal and final vertical components with Pythagoras, and find the angle with tan.

7 Method for an angled launch

1. Resolve: $u_H = u \cos\theta$, $u_V = u \sin\theta$.
2. Vertical: use suvat ($a = -9.81$ m s^{-2}, up +) to find t, or the maximum height ($v_V = 0$ at the top: $v^2 = u^2 + 2as$).
3. Horizontal: $s = u_H \times t$.
4. Compare with the target (both numbers, with units) and conclude.

WORKED EXAMPLE: ANGLED LAUNCH

A ball is kicked from the ground at 18 m s^{-1} at 35° above the horizontal. Find its maximum height, time of flight and range.
$u_H = 18 \cos 35° = 14.7$ m s^{-1}. $u_V = 18 \sin 35° = 10.3$ m s^{-1}.
Max height: $0 = 10.3^2 + 2(-9.81)H \rightarrow H = 10.3^2 \div 19.62 =$ **5.4 m**.
Time to top: $0 = 10.3 - 9.81t \rightarrow t = 1.05$ s. It lands at launch height, so time of flight = 2 × 1.05 = **2.1 s**.
Range = 14.7 × 2.10 = **31 m**.

WORKED EXAMPLE: HORIZONTAL LAUNCH

A marble rolls off a 1.2 m high table at 2.5 m s^{-1}. Where does it land, and how fast is it going?
Vertical ($u_V = 0$, down +): $1.2 = \frac{1}{2} \times 9.81 \times t^2 \rightarrow t = \sqrt{(2.4 \div 9.81)} = 0.495$ s.
Horizontal: $s = 2.5 \times 0.495 =$ **1.2 m** from the table.
Final vertical velocity = 9.81 × 0.495 = 4.85 m s^{-1}. Speed = $\sqrt{(2.5^2 + 4.85^2)} =$ **5.5 m s^{-1}**, at $\tan^{-1}(4.85 \div 2.5)$ = **63° below the horizontal**.

7 Effect of air resistance (a common explain question)

- Air resistance acts **against the velocity**, so it has a horizontal **and** a vertical component.
- The horizontal velocity **decreases** (the object decelerates horizontally), so the range is shorter.
- On the way up, the vertical deceleration is **more** than g, so the maximum height is lower. The path is no longer symmetrical: it comes down more steeply.

MARK-SCHEME LANGUAGE

- Final mark of a "will it clear / reach" question: "**Comparison of … with … and consistent conclusion**".
- "Use of $s = ut + \frac{1}{2}at^2$ with $a = 0$" for the horizontal motion is credited.
- Air resistance: "air resistance acts on the projectile; it decelerates horizontally; the vertical acceleration is no longer 9.81 m s^{-2}" (WPH11 Jan 2026 Q18).
- Path sketches: a smooth curve (parabola). "Do not accept a line that curves back towards the cliff" (WPH11 Jun 2026 Q18).
- Assumptions: say air resistance is "**negligible**" or "ignored", not "there is none".

Often asked as: launches from cliffs, platforms and ramps; "will the ball clear the wall / reach the platform?"; max height and range; the angle of the velocity at landing; 6-mark "how would air resistance change the motion?"; sketches of the path.

COMMON MISTAKE

Using a = 9.81 for the horizontal motion, or using the launch speed as the vertical velocity of a horizontal launch.
Doubling the time to the top when the landing height differs from the launch height. Use the quadratic, or time up + time down from the peak.
Giving the maximum height above the launch point when the question wants height above the ground (add the starting height).
Explaining horizontal motion with a "forward force". Nothing pushes it forward: no horizontal force means constant horizontal velocity.
Saying "the range is shorter" without saying **which component** changes and why.

BEYOND THE SPEC, BUT EXAMINED

- Landing on a slope or ramp rather than level ground (WPH11 Jan 2020 Q15).
- The **angle of the final velocity** from its components (e.g. 33° from the vertical, WPH11 Oct 2024 Q15).
- Strobe images of a projectile: equal horizontal spacing, increasing vertical spacing (WPH11 Jun 2019 Q18).

SPEC 8 Free-body force diagrams and centre of gravity

WHAT THIS SPEC POINT ASKS

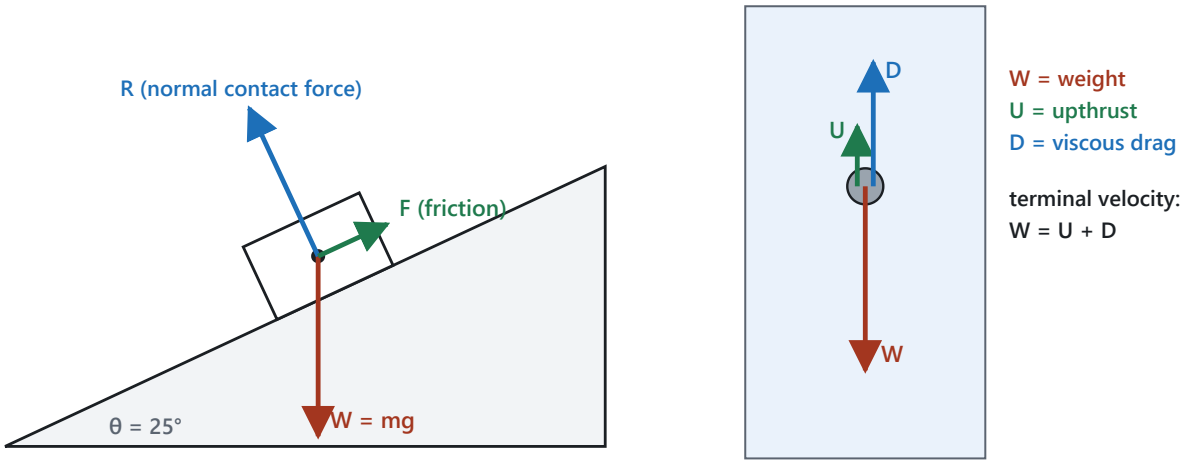
Draw and interpret free-body diagrams for a particle or for an extended rigid body, using the idea of the centre of gravity.

8 What a free-body diagram is

A diagram of **one object on its own**, showing every force **acting on it** (not forces it exerts on other things).

1. Draw the object as a dot (particle) or a simple shape (extended body).
2. Draw each force as an arrow **starting on the object** (on the dot), pointing the right way, drawn with a ruler.
3. Make arrow lengths show relative sizes. Forces that balance get equal lengths.
4. Label each arrow with its name: **weight (W or mg)**, **normal contact force (R or N)**, **friction (F)**, **tension (T)**, **drag / air resistance (D)**, **upthrust (U)**, thrust.

- **Weight** acts vertically down from the **centre of gravity** (16).
- The **normal contact force** is perpendicular to the surface.
- **Friction** acts along the surface, opposing (attempted) sliding. **Drag** acts opposite to the velocity.
- Only forces go on the diagram. Never draw "ma", velocity, acceleration, the resultant, or components (mg sin θ, mg cos θ) as extra forces.



(a) block at rest on a rough slope

(b) sphere falling through a liquid

(a) Three forces on a block at rest on a slope: they close into a triangle because the block is in equilibrium. (b) A sphere at terminal velocity in a liquid: weight = upthrust + drag, so the arrow lengths match.

8 Particle or extended body?

- **Particle**: all forces act through one point. Use it when only translation (moving along) matters.
- **Extended rigid body** (a beam, ladder, person): forces act at **different points**, so they can cause turning. Weight acts at the centre of gravity; supports act where they touch. This matters for moments (15, 16).
- **Three forces in equilibrium** (not parallel): drawn tip to tail they make a **closed triangle**, and their lines of action pass through one point.

MARK-SCHEME LANGUAGE

- "−1 for each extra force; −1 if any arrow does not touch the dot; −1 if any arrow is not close to vertical" (for a sphere in oil) (WPH11 Oct 2019 Q15).
- Lengths are often marked: "length of weight arrow ≈ length of drag + upthrust" (WPH11 Jan 2024 Q15); "weight and normal arrows equal and twice the forward-force arrow" (WPH11 Jan 2026 Q11).
- Weight labels accepted: "weight", "W", "mg". **Not "G".** "Do not allow more than three arrows" when three forces are expected.

Often asked as: forces on a sledge on a slope, a sphere in oil, a ball hanging in water, a train engine; adding forces to an extended body (weight at the centre of gravity, reaction at a pivot); MCQ "which force is wrongly drawn?".

COMMON MISTAKE

Adding a drag or friction arrow when the question says it is negligible: an extra arrow scored 0 (WPH11 Oct 2024 Q16).
Drawing a "forward force" for something moving at constant velocity with nothing pushing it.
Drag instead of upthrust for an object held still in water (WPH11 Jan 2020 Q19).
Freehand arrows, arrows not touching the object, unequal lengths for balanced forces.

BEYOND THE SPEC, BUT EXAMINED

- When forces come in pairs, name **what exerts the force on what**: "the Earth on the person" (WPH11 Oct 2020 Q13).
- Upthrust acts through the centre of gravity of the **displaced fluid**; as a boat tilts, this point moves, which can turn the boat back upright (WPH11 Oct 2019 Q14).

SPEC 9

Newton's first and second laws (F = ma) and terminal velocity

WHAT THIS SPEC POINT ASKS

Use ΣF = ma (Newton's second law, constant mass), including the first law case where a = 0 (at rest or constant velocity). Use the term terminal velocity.

9 Newton's first law

An object stays **at rest** or moves at **constant velocity** unless a **resultant force** acts on it. Resultant force = 0 ⇔ acceleration = 0. (Moving does not need a force; **changing** velocity does.)

9 Newton's second law

$$\Sigma F = ma$$

ΣF = resultant (net) force in N, m = mass in kg, a = acceleration in m s^{-2}. The acceleration is **in the direction of the resultant force**. 1 N is the force that gives 1 kg an acceleration of 1 m s^{-2}.

Method:

1. Free-body diagram.
2. Choose a positive direction (usually the direction of the acceleration).
3. ΣF = (forces in + direction) − (forces in − direction) = ma.

Standard set-ups from past papers:

Situation	Equation (+ = direction of acceleration)
Rocket lifting off	thrust − W = ma
Lift / trampoline / cable pulling up	T − W = ma
Sphere falling in a fluid	W − U − D = ma
Sliding down a slope with friction	mg sin θ − F = ma
Car on the level	driving force − resistive forces = ma

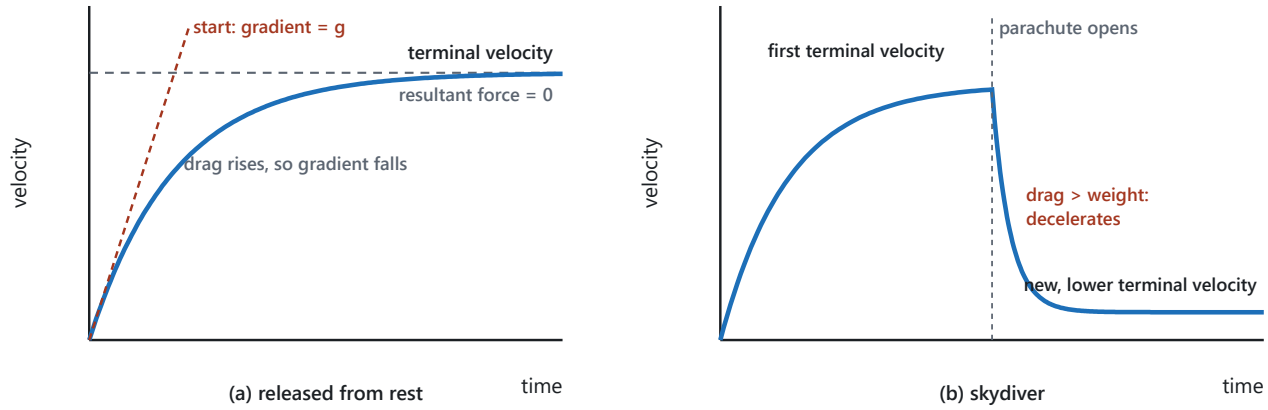
WORKED EXAMPLE: A CAR

A 1200 kg car has a driving force of 3000 N and total resistive forces of 900 N.
ΣF = 3000 − 900 = 2100 N. a = 2100 ÷ 1200 = **1.75 m s^{-2}** (1.8 m s^{-2} to 2 s.f.).

WORKED EXAMPLE: WHAT A SCALE READS IN A LIFT

A 70 kg person stands on scales in a lift accelerating **upwards** at 1.5 m s^{-2}. What do the scales read?
The scales read the **normal contact force** R, not the weight. Up +: R − mg = ma.
R = m(g + a) = 70 × (9.81 + 1.5) = **790 N** (their weight is still 70 × 9.81 = 687 N).
Accelerating downwards (or decelerating while going up), R < mg and the person "feels lighter".

9 Terminal velocity



(a) Released from rest: the gradient (acceleration) starts at g and falls to zero. (b) Opening the parachute makes drag bigger than weight: the skydiver decelerates to a new, lower terminal velocity.

Explain it as a chain, **in terms of forces**:

1. At release, v = 0 so drag = 0. The only force is weight (ignoring upthrust), so a = g.
2. As speed increases, **drag increases**, so the **resultant force** (W − D) decreases and the **acceleration decreases** (the speed still rises, just more slowly).
3. Eventually drag (+ upthrust) = weight. **Resultant force = 0, so acceleration = 0**, and the object moves at a constant **terminal velocity** (Newton's first law).

Parachute opening: drag suddenly increases (larger area), so **drag > weight**, the resultant force is **upwards** and the skydiver **decelerates**. As speed falls, drag falls, the deceleration decreases, until drag = weight again at a **lower** terminal velocity.

Vehicles have a "terminal velocity" too: a car reaches top speed when the driving force equals the resistive forces.

MARK-SCHEME LANGUAGE

- Terminal velocity chain (5 marks, WPH11 Jun 2025 Q20): "air resistance > weight → resultant force upwards → deceleration"; "as speed decreases, air resistance decreases, so deceleration decreases"; "until air resistance (+ upthrust) = weight, so resultant force = 0 and acceleration = 0".
- "Allow drag for air resistance." "Ignore 'forces become balanced'" and "ignore 'speed is constant'" unless the **resultant force** step is there.
- Terminal velocity: "constant (maximum) velocity when the resultant force is zero (W = D + U)".
- Resultant force: "the single force that has the same effect as all the forces acting".
- If the question says "**using Newton's laws**", name the law you use at each step.

Often asked as: ΣF = ma with two or three forces (rocket, trampolinist, lift bag, boat); "explain why the sphere reaches terminal velocity"; parachute opening; 6-mark force explanations (skydiver, lift and scales, bungee jumper, balloon).

COMMON MISTAKE

Treating thrust as the resultant force and forgetting the weight.
Explaining only the steady state, and not why the object accelerated first (or the reverse). Cover both stages.
"Decreasing acceleration means decreasing velocity." No: the velocity still increases, more slowly.
Answering in **energy** when the question says "in terms of forces".
Saying a person in a lift "weighs less". Their weight is unchanged; the **contact force** changes.
Quoting all three laws without applying them to the situation.

6-MARK LINKAGE ANSWERS

These are marked as up to 4 marks for indicative points and up to 2 for linking them into a logical chain. 6 points with clear reasoning = 6; 4 points = 4 at most.

- Tell the story **in time order**, one stage at a time ("at the start… as it speeds up… eventually…").
- For each force: name it, give its direction and what exerts it. Say "**resultant force**" and give its direction.
- Cover **every** stage the question asks about, and refer to the graph if there is one.

BEYOND THE SPEC, BUT EXAMINED

- A balance, bathroom scale or force plate reads the **normal contact force**, not the weight (WPH11 Jan 2022 Q14, Oct 2024 Q15).
- Changing mass: a rocket's acceleration increases as fuel burns, because both its weight and its mass fall (WPH11 Oct 2023 Q11).
- Force = rate of change of momentum, F = Δp/Δt (see 13) (WPH11 Jan 2021 Q14).

SPEC 10

Gravitational field strength and weight

WHAT THIS SPEC POINT ASKS

Use g = F/m and W = mg.

$$g = F \div m \qquad W = mg$$

- **Gravitational field strength** g = the gravitational force per unit mass (weight per unit mass). Unit **N kg^{-1}**. Near the Earth's surface, g = 9.81 N kg^{-1}.
- **Mass** (kg, scalar) is the amount of matter; it is the same everywhere. **Weight** (N, vector) is the gravitational force on the mass, acting towards the centre of the planet; it depends on g.
- N kg^{-1} = m s^{-2}. In **free fall** (weight the only force), a = W ÷ m = g, so every object falls with the same acceleration whatever its mass. A hammer and a feather dropped on the Moon land together.

WORKED EXAMPLE: WEIGHT ON THE MOON

An astronaut has mass 75 kg. g on the Moon = 1.63 N kg^{-1}.
W_{Moon} = 75 × 1.63 = **122 N**. W_{Earth} = 75 × 9.81 = **736 N**. The mass is still 75 kg.

MARK-SCHEME LANGUAGE

- "Use of W = mg" is a mark by itself. An answer in kg where a force was asked loses it.
- g defined as "**weight per unit mass**" or "force per unit mass".
- Free fall: "**weight is the only force acting**". This is not the same as terminal velocity.

Often asked as: MCQ definition of g; using g for Mars or the Moon (value given in the question); "is the probe in free fall?" (compare its acceleration with g); W = mg inside almost every dynamics, upthrust and energy question.

COMMON MISTAKE

Forgetting to multiply by g (using a mass in kg as a force).
Using g = 10 m s^{-2}: loses a mark.
Saying gravity "gets weaker" over the few metres of a lift ride or a jump. Near the surface, g is constant.

SPEC 11

Core Practical 1: acceleration of a freely falling object

WHAT THIS SPEC POINT ASKS

Determine the acceleration of free fall (g) by experiment.

11 Principle

Dropped from rest (u = 0): s = ½gt^2. So a graph of **s against t^2** is a straight line through the origin with **gradient = g/2**, i.e. **g = 2 × gradient**.

CORE PRACTICAL 1: METHOD IN BRIEF

Electromagnet and trapdoor (or two light gates)

1. Clamp an electromagnet holding a steel ball above a trapdoor switch. Switching off the magnet starts the timer; the ball hitting the trapdoor stops it.
2. Measure the drop height h from the **bottom of the ball** to the trapdoor with a metre rule (use a set square to avoid parallax; check vertical with a plumb line).
3. Release and record t. **Repeat 3 times** and take a mean.
4. Repeat for at least **5–6 heights** (e.g. 0.30 m to 1.20 m).
5. Plot h (y-axis) against t^2 (x-axis). Draw the best-fit line. g = 2 × gradient.

Other versions: a card of known length falling through a light gate (v = card length ÷ time; plot v^2 against h, gradient = 2g); a ball passing two light gates; strobe photography or video of a falling ball next

to a scale.
Safety: catch the ball in a tray of sand or on a cushion; clamp the stand firmly.

11 Variables and errors

- **IV**: drop height. **DV**: fall time. **Control**: same ball, released from rest each time.
- **Systematic error**: the magnet releases the ball a little late (residual magnetism), so every t is too long. On the graph this shows as an **intercept**; the gradient, and so g, is unaffected. That's one reason to use a graph.
- **Random error**: hand timing with a stopwatch involves **human reaction time**: large random errors. Use light gates or an electronic timer.
- Use **larger heights** to make t larger and its percentage uncertainty smaller. Air resistance is negligible for a small dense ball.

WORKED EXAMPLE: G AND ITS UNCERTAINTY

h = 0.800 m ± 0.002 m, mean t = 0.404 s ± 0.005 s (half the range of repeats).
$g = 2h \div t^2 = 2 \times 0.800 \div 0.404^2 =$ **$9.80\ m\ s^{-2}$**.
% uncertainty in h = 0.002 ÷ 0.800 × 100 = 0.25 %. % uncertainty in t = 0.005 ÷ 0.404 × 100 = 1.24 %.
Combining: add the percentage uncertainties, and **double** the one for t because t is squared: 0.25 + 2 × 1.24 = **2.7 %**.
g = 9.8 m s^{-2} ± 2.7 % (± 0.27 m s^{-2}): the accepted 9.81 m s^{-2} lies within this range.

MARK-SCHEME LANGUAGE

- "Reference to $s = ut + \frac{1}{2}at^2$ with u = 0"; "correct variables on axes to give a straight line through the origin (s against t^2)"; "units"; "g = 2 × gradient" (WPH11 Jan 2020 Q11).
- With one light gate and a falling rod: measure the rod's length and the distance from release to the gate; v = length ÷ time; repeat at other heights; plot v^2 against s, gradient = 2a (WPH11 Jan 2019 Q12).
- Stopwatch timing: "involves human reaction time, so (large) **random** errors". "Ignore human error" (WPH11 Oct 2025 Q13).
- Strobe photo: "$s = \frac{1}{2}gt^2$ and g is constant, compare with y = mx; gradient constant"; then "use of a pair of values or the gradient, g = …". A bare "g = 9.81" with no working scored 0 (WPH11 Oct 2024 Q14).

Often asked as: describe a method (5–7 marks); which graph to plot and how to get g from it; why manual timing is poor; why a shorter flash gives less blur; MCQ on what overestimating the distance does to g (too high).

COMMON MISTAKE

"The height" with no start and end point. Say "from the bottom of the ball to the trapdoor/light gate".
Plotting a graph that doesn't match the equation the question names (a v–t graph when it says $v^2 = u^2 + 2as$).
Forgetting to **double** the gradient of s against t^2.
Writing g with no unit, or using u = 0 without saying so.

SPEC 12

Newton's third law

WHAT THIS SPEC POINT ASKS

Know Newton's third law and the properties of the pair of forces in an interaction between two bodies.

12 The law

If body A exerts a force on body B, then B exerts a force on A that is **equal in size and opposite in direction**.

A third-law pair always:

- acts on **two different bodies** (one force on each);
- is the **same type** of force (both gravitational, both contact, both magnetic…);
- is **equal in magnitude** and **opposite in direction**;
- acts **at the same time** and along the same line.

12 Examples

Force	Its third-law partner
Earth pulls the apple down (weight)	apple pulls the Earth up (gravitational)
Foot pushes the ground backwards	ground pushes the foot **forwards** (this is what moves you)
Rocket pushes exhaust gas down	gas pushes the rocket up (thrust)
Ball in water: water pushes the ball up (upthrust)	ball pushes the water down, so a balance under the beaker reads more

12 First law or third law? The book on a table

A book rests on a table. Its weight W (Earth on book, gravitational) and the normal contact force R (table on book, contact) are equal and opposite. They are **not** a third-law pair: they act on the **same** body and are **different types**. They balance because the book is in equilibrium (**first law**). The partner of W is the book pulling the Earth up; the partner of R is the book pushing down on the table.

Why third-law forces don't cancel: they act on **different** objects, so they never appear on the same free-body diagram.

MARK-SCHEME LANGUAGE

- Not a pair because "the forces act on the **same object**" and "they are **different types** of force" (WPH11 Jun 2022 Q11).
- A pair: "equal in magnitude, opposite in direction, same type, act on different bodies".
- Applying it: "P exerts a force on Q, so Q exerts a force on P; by Newton's third law the forces are equal and opposite" (WPH11 Jun 2022 Q17).

Often asked as: "are these two forces a Newton's third law pair? Explain"; MCQs choosing the partner force; applications (shunting engine and rail, ball in water on a balance, racket and ball, gliders).

COMMON MISTAKE

Not saying **which object exerts the force and which object it acts on**. Write "the rail exerts a forward force on the engine".
Giving only "equal and opposite" without saying why the two named forces fail the test.
"Same but opposite": say **equal in size** and **opposite in direction**.
Mixing up the second and third laws. A third-law pair never adds to zero on one body.

SPEC 13

Momentum

WHAT THIS SPEC POINT ASKS

Understand that momentum is defined as p = mv.

$$p = mv$$

p = momentum (kg m s^{-1}, or N s), m = mass (kg), v = velocity (m s^{-1}).

- Momentum is a **vector**, in the direction of the velocity. In one dimension, use + and − signs.
- A slow heavy object and a fast light one can have the same momentum.

13 Momentum and force

Newton's second law can also be written as **force = rate of change of momentum** (beyond the spec wording, but examined):

$$F = \Delta p \div \Delta t = (mv - mu) \div t$$

With constant mass this is just F = ma. The product FΔt is the **change in momentum** (sometimes called impulse). Spreading the change over a longer time means a smaller force: crumple zones, airbags, bending your knees on landing.

WORKED EXAMPLE: MOMENTUM AND A REBOUND

A cricket ball of mass 0.16 kg moves at 35 m s^{-1}: p = 0.16 × 35 = **5.6 kg m s^{-1}**.
A 0.060 kg ball hits a wall at 20 m s^{-1} and rebounds at 15 m s^{-1}. Take towards the wall as +.
Δp = m(v − u) = 0.060 × (−15 − 20) = **−2.1 kg m s^{-1}** (the change is away from the wall).
If contact lasts 0.050 s, the average force on the ball = 2.1 ÷ 0.050 = **42 N**, away from the wall.

MARK-SCHEME LANGUAGE

- "Use of p = mv" is a mark in nearly every collision question. A momentum or velocity answer **without a unit** loses the last mark (WPH11 Jan 2025 Q14).
- Units: kg m s^{-1} or N s. kg m s^{-2} is the unit of **rate of change** of momentum (= N) (WPH11 Jan 2021 Q14).

Often asked as: p = mv for an object; total momentum of a system; MCQs on the unit; F = Δp/Δt for a collision or a rocket.

COMMON MISTAKE

Confusing momentum (mv) with kinetic energy (½mv^2).
Ignoring direction in a rebound: the change is 20 − (−15) = 35, not 20 − 15 = 5.
Leaving mass in grams.

SPEC 14

Conservation of linear momentum

WHAT THIS SPEC POINT ASKS

Know the principle of conservation of linear momentum, relate it to Newton's laws, and apply it to problems in one dimension.

14 The principle

The **total momentum of a system stays constant, provided no external (resultant) force acts on the system** (a closed system). So:

$$\text{total momentum before} = \text{total momentum after}$$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

14 Link to Newton's laws

When A and B collide, A pushes B and B pushes A with **equal and opposite** forces (third law), for the **same time**. So FΔt (the change in momentum) is equal and opposite for each body (second law as F = Δp/Δt). One gains exactly what the other loses: the total is unchanged. The forces between the two are internal; only an **external** force could change the total.

14 Collisions and explosions

- **Collision, sticking together**: $m_1u_1 + m_2u_2 = (m_1 + m_2)v$.
- **Explosion / recoil** from rest: total momentum before = 0, so after, the two pieces move in **opposite directions** with equal-sized momenta (a gun recoils; a skateboarder throwing a ball rolls backwards).
- Momentum is **always** conserved in a closed system. **Kinetic energy is not necessarily conserved**:
  - **elastic** collision: total KE the same before and after;
  - **inelastic** collision: some KE is transferred to heat, sound and deforming the objects. Objects that stick together always lose KE;
  - in an explosion, KE **increases** (it comes from a chemical or elastic store).

WORKED EXAMPLE: TROLLEYS THAT STICK

A 0.80 kg trolley at 3.0 m s^{-1} hits a stationary 1.60 kg trolley; they stick together.
Before: 0.80 × 3.0 + 1.60 × 0 = 2.4 kg m s^{-1}. After: (0.80 + 1.60)v = 2.4 → v = **1.0 m s^{-1}** in the original direction.
KE before = ½ × 0.80 × 3.0^2 = 3.6 J. KE after = ½ × 2.40 × 1.0^2 = 1.2 J. KE is lost, so the collision is **inelastic**.

WORKED EXAMPLE: AN EXPLOSION (SPRING-LOADED TROLLEYS)

Two trolleys of mass 0.50 kg and 0.75 kg are pushed apart by a spring from rest. The 0.50 kg trolley moves **left** at 0.60 m s^{-1}. Take right as +.
0 = 0.50 × (−0.60) + 0.75v → v = 0.30 ÷ 0.75 = **0.40 m s^{-1} to the right**.

14 Method

1. Sketch before and after, with masses and velocity arrows. **Choose + direction.**
2. Write total momentum before = total after, with signs.
3. Solve. A **negative** answer means the object moves in the − direction: say so.
4. If you're given **kinetic energy**, find v first from $E_k = ½mv^2$, then p = mv.

MARK-SCHEME LANGUAGE

- Principle (2 marks): "the **total** momentum (of a system) is constant / total before = total after"; "**provided no external forces act** / in a closed system" (WPH11 Oct 2025 Q14, Jan 2019 Q15).
- "A negative sign is sufficient to show the direction reversed", but you may be asked to **state** what it means (WPH11 Jan 2023 Q16).
- Showing momentum is conserved: calculate the total **before** and **after** with numbers (WPH11 Oct 2023 Q12). Repeating the stem scores nothing.
- Two credited explanations (WPH11 Jun 2026 Q13): momentum conserved, so equal and opposite changes in momentum; **or** third-law forces equal and opposite for the same time, so equal Δp. The heavier object has the smaller change in velocity.

Often asked as: state the principle; collisions and explosions (trolleys, gliders, a boat ejecting steam, a skater throwing a ball); KE lost in a collision; "explain in terms of momentum why…".

COMMON MISTAKE

Leaving out "**total**" or "**sum**" from the statement. It is the most common lost mark.
Stating the condition without applying it to the named system ("no external force acts on the two cars").
Giving both objects positive momentum when they move in opposite directions.
Assuming objects stick together when the question doesn't say so.
Equating the GPE of one trolley to the KE of two after a collision: momentum must be used for the collision itself (WPH11 Oct 2019 Q12).
Answering in forces when the question says "in terms of momentum" (or the reverse).

BEYOND THE SPEC, BUT EXAMINED

- Impulse route: average force = Δp ÷ Δt (WPH11 Jun 2022 Q17).
- Rocket thrust = rate of change of momentum of the exhaust gases (WPH11 Jan 2023 Q16).
- KE before and after: a perfectly inelastic collision always loses KE (WPH11 Jun 2021 Q11).

SPEC 15

Moment of a force

WHAT THIS SPEC POINT ASKS

Use moment of a force = Fx, where x is the perpendicular distance from the axis of rotation to the line of action of the force.

$$\text{moment} = F \times x$$

F = force (N), x = **perpendicular** distance from the pivot to the **line of action** of the force (m). Unit: **N m** (N cm if you work in cm, but convert for final answers).

- A moment is a turning effect, clockwise or anticlockwise.
- Bigger force or bigger perpendicular distance → bigger moment. That's why door handles are far from the hinge and long spanners help.
- If the force is at angle θ to a bar of length d, the perpendicular distance is d sin θ. Or resolve the force: only the component perpendicular to the bar turns it. A force whose line of action passes through the pivot has **zero** moment.

WORKED EXAMPLE: AN ANGLED FORCE

A 40 N force acts on the end of a 0.25 m spanner at 60° to the spanner.
Perpendicular distance = 0.25 sin 60° = 0.217 m. Moment = 40 × 0.217 = **8.7 N m**.
Same answer by resolving: perpendicular component = 40 sin 60° = 34.6 N; 34.6 × 0.25 = 8.7 N m.

MARK-SCHEME LANGUAGE

- "Use of moment = Fx"; "use of trigonometry for the perpendicular distance / component".
- Unit **N m** (not N m^{-1}, not N) (WPH11 Oct 2024 Q13).
- Why a moment falls as the angle changes: "the perpendicular distance (x cos θ) decreases".

Often asked as: moment with an angled force; why a moment changes as something rotates; unit MCQs; the moment a motor must provide.

COMMON MISTAKE

Using the distance along the bar instead of the **perpendicular** distance.
Unit N m^{-1}. Moments are in N m (don't write J, even though the base units match).

BEYOND THE SPEC, BUT EXAMINED

- **Couple**: two equal, opposite, parallel forces not in line. No resultant force, but a turning effect: moment of a couple = one force × perpendicular distance between them (steering wheel, tap).
- **Toppling**: an object topples when its centre of gravity moves beyond the edge of its base, because the weight then gives a moment that turns it further over. Low centre of gravity and a wide base = stable (WPH11 Oct 2021 Q19).
- Estimating a length you aren't given (a forearm is 30–50 cm) and using it (WPH11 Jun 2019 Q15).

SPEC 16

Centre of gravity and the principle of moments

WHAT THIS SPEC POINT ASKS

Use the centre of gravity of an extended body and apply the principle of moments to an extended body in equilibrium.

16 Centre of gravity

The **centre of gravity** is the point through which the **weight** of an object **appears to act**. Every particle of the object has weight; the centre of gravity is where one single total weight can be drawn with the same effect.

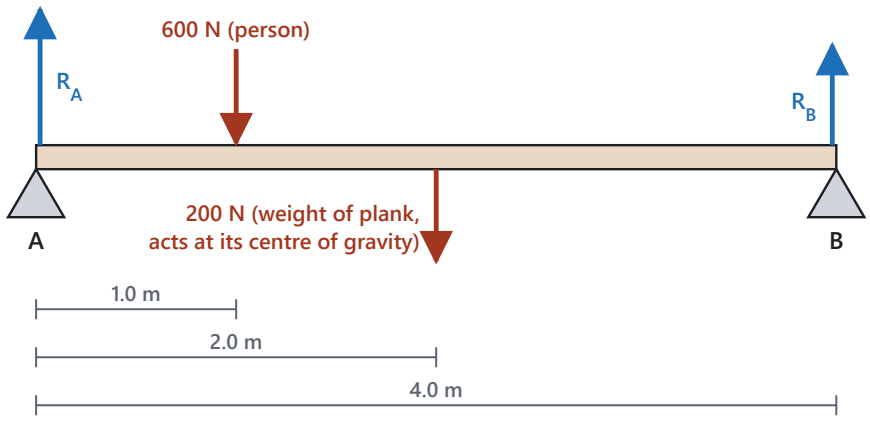
- For a **uniform** object (rod, beam, plank) it's at the geometric centre.
- **Irregular rod**: balance it on a knife edge; the balance point is the centre of gravity.
- **Irregular flat shape (lamina)**: hang it from a pin with a plumb line, draw the vertical line, repeat from a second hole; the lines cross at the centre of gravity.

16 Equilibrium

An object is in equilibrium when there is:

1. **no resultant force** (forces up = forces down, forces left = forces right), **and**
2. **no resultant moment** about any point.

Principle of moments: for an object in equilibrium, the **total** clockwise moment about any point = the **total** anticlockwise moment about that point.



A uniform 4.0 m plank (weight 200 N) on supports A and B, with a 600 N person 1.0 m from A.

WORKED EXAMPLE: TWO SUPPORTS

Find the support forces R_A and R_B for the plank in the figure.
Take moments about A (so R_A has no moment).
Clockwise: 600 × 1.0 + 200 × 2.0 = 1000 N m. Anticlockwise: R_B × 4.0.
R_B × 4.0 = 1000 → R_B = **250 N**.
No resultant force: R_A + R_B = 600 + 200 → R_A = **550 N**.
Check with moments about B: 550 × 4.0 = 600 × 3.0 + 200 × 2.0 = 2200 N m. R_A is bigger because the person is nearer A.

16 Method

1. Draw every force on the diagram, **including the object's own weight at its centre of gravity** and **both** support forces. Mark the distances.
2. **State the pivot** you take moments about. Choose a point where an unknown force acts.
3. Total clockwise moment = total anticlockwise moment.
4. Use "up = down" for the second unknown.
5. Angled forces: resolve them, or find the perpendicular distance with trig.

MARK-SCHEME LANGUAGE

- Centre of gravity: "the point through which the weight (of the object) **appears to act**" / "may be taken to act" (WPH11 Jun 2026 Q16, Oct 2023 Q15).
- Equilibrium (2 marks): "**no resultant force**"; "**no resultant moment** (about any point)" (WPH11 Jan 2024 Q18). "Clockwise moment = anticlockwise moment" is **not enough**: it must say **total** or sum of moments (WPH11 Oct 2025 Q20).
- Calculation marks: "use of moment = Fx", "use of the principle of moments", "use of ΣF = 0 for the other support", then comparison and conclusion.
- Drawing the weight at the centre of gravity is a shortcut: the weight "does not really act at one point" (WPH11 Jan 2023 Q13).

Often asked as: define centre of gravity; state the two conditions for equilibrium; shelves, cranes, bridges, arms and doors with one or two unknown forces; explain why a force changes when a load moves; toppling.

COMMON MISTAKE

"The weight acts at the centre of gravity" or "all the mass is concentrated there": the word **appears** is the mark.
Giving only the moments condition and forgetting "no resultant force" (or the reverse).
Forgetting the beam's own weight or the second support force; not stating the pivot (a lost mark in WPH11 Oct 2022 Q13).
Using the full width of a door as the distance to its centre of gravity instead of half (WPH11 Jan 2024 Q18).
"Forces are balanced" on its own adds nothing.

BEYOND THE SPEC, BUT EXAMINED

- Model limits: an arm is not a uniform beam, so its centre of gravity isn't at its middle (WPH11 Jun 2019 Q15).
- Comparing a calculated moment with the most a fixing can take, and concluding (WPH11 Jun 2022 Q13).
- A book pushed over a shelf edge tips when its centre of gravity passes the edge (WPH11 Jan 2020 Q13).

SPEC 17

Work done

WHAT THIS SPEC POINT ASKS

Use $\Delta W = F\Delta s$, including when the force is not along the line of motion.

$$\Delta W = F\Delta s$$

ΔW = work done (J), F = force (N), Δs = distance moved **in the direction of the force** (m). 1 J is the work done when a 1 N force moves its point of application 1 m in the direction of the force.

- **Work done = energy transferred.** When a force does work on an object, energy is transferred to (or from) it.
- **Force at angle θ to the motion**: only the component along the motion does work.

$$\Delta W = F\Delta s \cos\theta$$

- Force **perpendicular** to the motion ($\theta = 90°$): **no work** (e.g. the normal contact force on a car on a level road).
- Work done **against** friction or air resistance is transferred to the surroundings as heat (thermal energy) and sound.

WORKED EXAMPLE: AN ANGLED PULL

A sledge is pulled 5.0 m along level snow by a rope with tension 30 N at 40° above the horizontal.
$\Delta W = 30 \times 5.0 \times \cos 40° =$ **115 J** (1.1×10^2 J to 2 s.f.).

WORKED EXAMPLE: UP A SLOPE

A box is pushed 6.0 m up a ramp by a force of 120 N acting **along** the ramp.
$\Delta W = 120 \times 6.0 =$ **720 J**. The force is along the motion, so no cos θ.

MARK-SCHEME LANGUAGE

- "Use of ΔW = FΔs" / "work done = force × distance moved **in the direction of the force**".
- The word **work** is needed in "explain" answers: "work is done against air resistance", "energy is dissipated".
- Work against a resistive force = total work in − energy gained (e.g. total work − final KE) (WPH11 Jun 2023 Q13).

Often asked as: MCQs with an angled force; work done by a boat or car in a given time; "explain why work is done"; work against air resistance or friction from an energy difference.

COMMON MISTAKE

Multiplying by cos θ or sin θ when the force is already **along** the motion (WPH11 Oct 2022 Q19). Answering in energy-store language only ("energy is transferred") when the question is about **work**. Using F = ma to find a force that isn't constant: use energy instead.

BEYOND THE SPEC, BUT EXAMINED

- Area under a force–distance graph = work done (same idea as elastic strain energy, 32).
- Deciding **which** force does the work: a falling bungee jumper's weight does work stretching the cord (WPH11 Jan 2021 Q15).

SPEC 18

Kinetic energy

WHAT THIS SPEC POINT ASKS

Use $E_k = ½mv^2$.

$$E_k = ½mv^2$$

E_k = kinetic energy (J), m = mass (kg), v = speed (m s^{-1}).

- Only v is squared. Doubling the speed **quadruples** the KE; doubling the KE multiplies the speed by √2.
- KE is a scalar: no direction, never negative.
- Where it comes from: a resultant force F accelerates a mass from rest over distance s. Work done = Fs = (ma)s, and $v^2 = 2as$, so $Fs = ½mv^2$.
- Braking distance ∝ v^2 (same braking force), because the work done by the brakes, Fs, must equal $½mv^2$.

WORKED EXAMPLE

A 1200 kg car at 15 m s^{-1}: $E_k = ½ \times 1200 \times 15^2$ = **1.35×10^5 J** (135 kJ).
Reverse: if $E_k = 1.2 \times 10^5$ J, $v = \sqrt{(2E_k \div m)} = \sqrt{(2 \times 1.2 \times 10^5 \div 1200)}$ = **14 m s^{-1}**.

MARK-SCHEME LANGUAGE

- "Use of $E_k = ½mv^2$". In "show that" questions give one more s.f. than printed.

Often asked as: KE of a moving object; speed from a given KE (e.g. 120 kJ), then momentum; ratio MCQs (2 × KE → v × √2).

COMMON MISTAKE

Forgetting to square v (a calculator slip examiners mention often).
Using the wrong mass (e.g. the whole system's when only one part moves).
Mixing up ½mv² and mv.

BEYOND THE SPEC, BUT EXAMINED

- Combining KE and momentum: given a KE, find v, then p = mv (WPH11 Oct 2025 Q14, Jan 2026 Q13). Equivalently, $E_k = p^2 \div 2m$.

SPEC 19 Gravitational potential energy

WHAT THIS SPEC POINT ASKS

Use $\Delta E_{grav} = mg\Delta h$ for changes in gravitational potential energy near the Earth's surface.

$$\Delta E_{grav} = mg\Delta h$$

ΔE_{grav} = change in gravitational potential energy (J), m = mass (kg), g = 9.81 N kg^{-1}, Δh = change in **vertical** height (m).

- It equals the work done against gravity when lifting: force mg × vertical distance Δh.
- Only valid **near the surface**, where g is constant.
- It depends only on the change in **vertical** height, not on the path. On a slope of length L at angle θ, Δh = L sin θ.

WORKED EXAMPLE

A 65 kg climber climbs a cliff 450 m high: $\Delta E_{grav} = 65 \times 9.81 \times 450 =$ **2.9×10^5 J**.
A child slides 2.5 m down a slide at 22° to the horizontal: Δh = 2.5 sin 22° = **0.94 m**, not 2.5 m.

MARK-SCHEME LANGUAGE

- "Use of $\Delta E_{grav} = mg\Delta h$"; choosing Δh as the **vertical** height is a separate mark.

Often asked as: GPE gained by counterweights, people, water; height from a slope length; nearly always inside an energy-conservation question.

COMMON MISTAKE

Using the distance **along a slope** as Δh.
Forgetting g, or using a mass where a weight is needed.
Adding KE when the question asks only for the GPE gained.

SPEC 20 Conservation of energy

WHAT THIS SPEC POINT ASKS

Know and apply the principle of conservation of energy, including work done, gravitational potential energy and kinetic energy.

20 The principle

Energy cannot be created or destroyed, only **transferred** from one store to another (or to the surroundings). The total energy of a closed system is constant.

20 Using it

- **No losses** (air resistance and friction negligible): GPE lost = KE gained.

$$mg\Delta h = \tfrac{1}{2}mv^2 \qquad \text{so} \qquad v = \sqrt{(2g\Delta h)}$$

The mass cancels: all objects reach the same speed after falling the same height.

- **With a resistive force**: the work done against it is transferred to the surroundings (heating).

$$\text{energy at start} = \text{energy at end} + \text{work done against resistive forces}$$

For example, GPE lost = KE gained + F × s (F = friction or drag, s = distance travelled).

- **Several stores**: elastic strain energy (32), GPE and KE can all appear, e.g. a pinball launcher: $\tfrac{1}{2}kx^2 \rightarrow \tfrac{1}{2}mv^2$.

WORKED EXAMPLE: SLEDGE WITH FRICTION

A 70 kg sledge starts from rest and slides 40 m down a slope at 15° to the horizontal. The friction force is 50 N. Find its speed at the bottom.
$\Delta h = 40 \sin 15° = 10.35$ m. GPE lost = 70 × 9.81 × 10.35 = 7110 J.
Work against friction = 50 × 40 = 2000 J.
KE at the bottom = 7110 − 2000 = 5110 J. v = √(2 × 5110 ÷ 70) = **12 m s^{-1}**.
You could also use ΣF = ma then suvat: both routes are credited.

20 When suvat can't be used

On a curved slide, a pendulum or a roller coaster, the acceleration **changes**, so the suvat equations don't apply. Energy still works, because it only needs the start and end states.

MARK-SCHEME LANGUAGE

- "Applies conservation of energy"; "E_{grav} decreases, E_k increases, and work is done against air resistance".
- At terminal velocity: all the GPE lost is transferred to the surroundings by work done against drag (WPH11 Jun 2025 Q20).
- "Ignore references to heating / thermal energy" on their own; "allow energy transferred to the surroundings due to work done against air resistance".
- A small mismatch between a printed "show that" value and the scheme's working (0.95 against 0.96 m s^{-1} in WPH11 Jan 2026 Q19) means either value is accepted.

Often asked as: speed at the bottom of a slope with or without friction; missing energy = work done against air resistance; spring or elastic energy → KE; 6-mark energy chains (falling, colliding, rising); why the equations of motion can't be used.

COMMON MISTAKE

Treating a collision with energy alone. Use **momentum** for the collision, and energy before and after it.
Including friction when the question says to ignore it, or ignoring it when it is given.
Saying energy is "lost". Say it is **transferred to the surroundings** (as heat) by work done against friction or drag.
Treating an object as falling freely when something else is pulling it.

BEYOND THE SPEC, BUT EXAMINED

- Choosing between the energy route and the force + suvat route: both score (WPH11 Jan 2019 Q18, Oct 2020 Q18).
- Elastic energy in conservation questions: $\frac{1}{2}kx^2 = \frac{1}{2}mv^2$ for a spring launcher (WPH11 Jan 2022 Q16), or $E_{el} + E_{grav} \rightarrow E_k$ (WPH11 Jan 2026 Q19).

SPEC 21

Power

WHAT THIS SPEC POINT ASKS

Use P = E/t and P = W/t.

$$P = E \div t \qquad P = W \div t$$

P = power (W), E = energy transferred (J), W = work done (J), t = time (s). **Power is the rate of energy transfer** (or rate of doing work). 1 W = 1 J s^{-1}.

- Watch prefixes and times: 1 kW = 10^3 W, 1 MW = 10^6 W. Convert minutes and hours to **seconds**. 1 kWh = 1000 W × 3600 s = 3.6 × 10^6 J.

21 P = Fv

For a force F moving something at velocity v (beyond the spec wording, but expected):

$$P = W \div t = F \times s \div t = Fv$$

A car at top speed: driving force = resistive force, so the engine's useful power = resistive force × speed. At constant power, the driving force falls as speed rises.

WORKED EXAMPLE: A CRANE

A crane lifts a 500 kg load through 12 m in 20 s at steady speed.
Work done = mgΔh = 500 × 9.81 × 12 = 58 900 J. P = 58 900 ÷ 20 = **2.9 kW** (2940 W).

WORKED EXAMPLE: P = FV

A car travels at a constant 30 m s^{-1} against resistive forces of 600 N.
Driving force = 600 N (constant velocity). P = Fv = 600 × 30 = **18 kW**.

MARK-SCHEME LANGUAGE

- "Use of P = W/t" / "use of P = E/t".
- Students who "know the composite formula P = Fv or derive it" (W = Fs, P = W/t, v = s/t) were credited (WPH11 Oct 2021 Q18).
- Power = "rate of energy transfer" or "rate of doing work" (only about a third chose this in an MCQ, WPH11 Oct 2019 Q8).

Often asked as: power of a lift, crane, motor or person; time from energy and power; why driving force falls as a car speeds up at constant power; power–time graphs; unit MCQs (W = J s^{-1} = kg m^2 s^{-3}).

COMMON MISTAKE

Hours to seconds: × 60 × 60, not × 60 once.
"Constant power means constant force." No: at constant power, F = P/v falls as v rises.
Resolving a force along a slope when it was already given along the slope (WPH11 Oct 2021 Q18).

SPEC 22

Efficiency

WHAT THIS SPEC POINT ASKS

Use efficiency = useful energy output ÷ total energy input, and the same with power.

efficiency = useful energy output ÷ total energy input

efficiency = useful power output ÷ total power input

- Efficiency has **no unit**. Give it as a decimal (0.74) or a percentage (74 %, with the % sign).
- It can never be more than 1 (100 %). If you get more, you've divided the wrong way round.
- Wasted energy = total input − useful output = (1 − efficiency) × input.
- What counts as "useful" depends on the purpose: heat from a light bulb is wasted, heat from a heater is useful.
- Improving efficiency: reduce friction (lubricate), reduce air resistance (streamline), reduce heat loss (insulate).

WORKED EXAMPLE: FORWARDS AND BACKWARDS

The crane in 21 has a useful power output of 2940 W and its motor takes in 4.0 kW.
Efficiency = 2940 ÷ 4000 = 0.74 = **74 %**.
Reverse: a motor of efficiency 0.74 must deliver 2940 W. Input = output ÷ efficiency = 2940 ÷ 0.74 = **4.0 kW**. Divide; don't multiply.

MARK-SCHEME LANGUAGE

- "Use of efficiency = useful energy output ÷ total energy input" (or power).
- "Efficiency = 28 % (must include the % symbol)" (WPH11 Jun 2025 Q14).
- Scripts that label **input** and **output** clearly score best.
- One published scheme writes 73.9 kJ as "73.9 × 10^4 J" (WPH11 Jun 2022 Q14). It is 7.39 × 10^4 J.

Often asked as: efficiency of a motor, power station, regenerative brakes, or a pulley-and-ramp system; finding the input needed from the output and efficiency; wasted energy; MCQs where an option above 100 % must be rejected.

COMMON MISTAKE

Multiplying by the efficiency when you need to **divide** (finding the input from the output).
Adding wasted power to a total input that already includes it.
Using the wrong distance for the wrong mass in a pulley system (WPH11 Jun 2019 Q12).

Topic: Materials

SPEC 23 Density

WHAT THIS SPEC POINT ASKS

Use ρ = m/V.

$$\rho = m \div V$$

ρ = density (kg m^{-3}), m = mass (kg), V = volume (m^3).

- Density is mass per unit volume. It's a property of the **material**, not of the size of the object.
- Water ≈ 1000 kg m^{-3} (1.00 g cm^{-3}); steel ≈ 7800 kg m^{-3}; air ≈ 1.2 kg m^{-3}.
- **Volume formulas you must know** (they are not given): sphere V = $\frac{4}{3}\pi r^3$; cylinder V = $\pi r^2 h$; cuboid V = l × w × h. Cross-sectional area of a wire A = $\pi r^2 = \pi d^2/4$.
- **Unit conversions**: 1 cm^3 = 10^{-6} m^3; 1 mm^3 = 10^{-9} m^3; 1 g cm^{-3} = 1000 kg m^{-3}. The power applies to the whole conversion factor: (10^{-2} m)^3 = 10^{-6} m^3.
- Measuring density: regular solid → mass on a balance, dimensions with vernier calipers or a micrometer; irregular solid → volume by displacement in a measuring cylinder; liquid → mass of a measured volume (balance and measuring cylinder).

WORKED EXAMPLE: A STEEL BALL

A steel ball bearing (ρ = 7800 kg m^{-3}) has a diameter of 6.0 mm. Find its mass and weight.
r = 3.0 mm = 3.0 × 10^{-3} m. V = $\frac{4}{3}$ × π × (3.0 × 10^{-3})^3 = 1.13 × 10^{-7} m^3.
m = ρV = 7800 × 1.13 × 10^{-7} = **8.8 × 10^{-4} kg**. W = mg = **8.7 × 10^{-3} N**.

MARK-SCHEME LANGUAGE

- "Calculates volume of sphere / cylinder"; "use of ρ = m/V"; "use of W = mg" (separate marks).
- Examiners say the sphere and cylinder volume formulas are ones "candidates are expected to know" (WPH11 Jun 2022 Q19).

Often asked as: mass or weight of a sphere, cylinder or drop; density from given data; MCQ scaling (a cube with sides × 1.5 has 1.5^3 times the volume, but the **same density** if it's the same material).

COMMON MISTAKE

Using the diameter as the radius; forgetting to cube (or square) it; using the surface-area formula for the volume.
Stopping at the mass when the weight was needed.
Not noticing an object is hollow (WPH11 Jan 2022 Q15).
cm^3 → m^3 by dividing by 100 instead of 10^6.

SPEC 24 Upthrust

WHAT THIS SPEC POINT ASKS

Use the relationship upthrust = weight of fluid displaced.

24 Archimedes' principle

An object in a fluid (liquid or gas) feels an upward force, **upthrust**, equal to the **weight of the fluid it displaces**.

$$U = \text{weight of fluid displaced} = \rho_{\text{fluid}} \times V_{\text{displaced}} \times g$$

- **Why** it exists: pressure in a fluid increases with depth, so the fluid pushes up on the bottom of the object harder than it pushes down on the top. The difference is the upthrust.
- For a fully submerged object, $V_{\text{displaced}}$ = the object's volume. For a floating object, it's only the submerged part.
- Upthrust does **not** depend on how deep the object is (once fully submerged), or on what it is made of.

24 Floating and sinking

- **Floating** (at rest): **upthrust = weight** (not greater). A floating object displaces its own weight of fluid.
- **Sinking**: weight > maximum upthrust (when fully submerged). An object denser than the fluid sinks; this is **why** density matters, but explain it with forces.
- Resultant force on a submerged object = W − U (downwards if positive).
- In a less dense fluid (e.g. fresh water rather than seawater, or warmer liquid), a floating object must displace a **larger volume** to get the same upthrust, so it sits lower.

WORKED EXAMPLE: UPTHRUST ON A BALL

The 6.0 mm steel ball from 23 ($V = 1.13 \times 10^{-7}$ m^3) is fully submerged in oil of density 920 kg m^{-3}.
$U = 920 \times 1.13 \times 10^{-7} \times 9.81 =$ **1.0×10^{-3} N**. Its weight is 8.7×10^{-3} N, so it sinks.

WORKED EXAMPLE: AN ICEBERG

Ice has density 917 kg m^{-3} and seawater 1025 kg m^{-3}. What fraction of an iceberg is above the water?
Floating: U = W, so $\rho_{\text{sea}} V_{\text{sub}} g = \rho_{\text{ice}} V g \rightarrow V_{\text{sub}} \div V = 917 \div 1025 = 0.895$.
Fraction above the surface = 1 − 0.895 = **0.11** (about 10 %).

MARK-SCHEME LANGUAGE

- Definition: "an upwards force caused by the displacement of fluid / equal to the weight of fluid displaced" (WPH11 Oct 2023 Q17: only a small proportion scored).
- "Use of upthrust = weight of fluid displaced" (often with "use of ρ = m/V"); "weight of displaced water", not "water **replaced**".
- Name **which** weight: "the weight of the hydrometer", "the weight of seawater displaced" (WPH11 Oct 2024 Q16).

Often asked as: upthrust on balloons, bubbles, submarines and lifting bags; icebergs; 6-mark explanations (Galilean thermometer: temperature rises → liquid density falls → upthrust falls → bulb sinks; an egg in water); hydrometers; a ball held in water on a balance.

COMMON MISTAKE

Saying a floating object has U > W. Floating at rest: **U = W**.
Explaining only with densities ("it's less dense so it floats"): no marks unless you give the **force** argument.
Bringing viscosity into a static floating question.
Believing the water pushes down harder at depth, so the upthrust changes with depth.

BEYOND THE SPEC, BUT EXAMINED

- Hydrometers and Galilean thermometers: the same object floats at different levels in liquids of different density (WPH11 Jun 2022 Q15, Oct 2024 Q16).
- A Cartesian diver: squeezing the bottle compresses the air bubble, the displaced volume falls, upthrust falls, so the resultant force is downwards and it sinks (WPH11 Jan 2024 Q13).
- Newton's third law: the fluid pushes up on the object, so the object pushes down on the fluid; a balance under the beaker reads more (WPH11 Jan 2020 Q19).

SPEC 25 Viscous drag and Stokes' law

WHAT THIS SPEC POINT ASKS

(a) Use Stokes' law, F = 6πηrv. (b) Know that it only applies to small spheres moving slowly with laminar flow, and that viscosity depends on temperature.

25 Viscosity and flow

- **Viscosity** η is a measure of a fluid's resistance to flow (how "thick" it is). Unit: **Pa s** (= N s m^{-2} = kg m^{-1} s^{-1}). Water ≈ 1 × 10^{-3} Pa s; glycerol ≈ 1.4 Pa s at 20 °C.
- **Laminar (streamline) flow**: the fluid moves in smooth layers that don't mix; the velocity at any point is steady. Streamlines are smooth and don't cross.
- **Turbulent flow**: the layers break up into eddies and swirls; the velocity at a point keeps changing. It happens at high speeds and around large or unstreamlined objects.
- **Viscous drag** is the resistive force on an object moving through a fluid.

25 Stokes' law

$$F = 6\pi\eta r v$$

F = viscous drag (N), η = viscosity of the fluid (Pa s), r = radius of the sphere (m), v = speed of the sphere relative to the fluid (m s^{-1}).

It applies only when:

- the object is a **small sphere**;
- it moves at **low speed**;
- the flow is **laminar** (no turbulence).

25 Temperature

The viscosity of a **liquid decreases** as the temperature rises (syrup and oil run faster when warm). So a ball falls faster in warm oil. (Gases do the opposite, but that's not examined.) Always control or measure the temperature in a viscosity experiment.

25 Terminal velocity of a sphere in a liquid

The forces are weight W (down), upthrust U and drag F (both up). See the free-body diagram in 8 and the v–t graph in 9. At terminal velocity there is no resultant force:

$$W = U + F \qquad \text{so} \qquad F = W - U$$

Scaling: W and U ∝ r^3 but drag ∝ r, so a **bigger sphere** of the same material has a **higher** terminal velocity (v ∝ r^2).

25 Typical set-up (as in most papers)

1. $V = \frac{4}{3}\pi r^3$; $W = \rho_{sphere} V g$.
2. $U = \rho_{fluid} V g$.
3. At terminal velocity, $F = W - U$.
4. $F = 6\pi\eta r v$: solve for η or v, **or** compare the drag Stokes' law predicts with $W - U$ to decide whether the law applies.

WORKED EXAMPLE: TERMINAL VELOCITY IN GLYCEROL

A steel ball of radius 1.0 mm (ρ = 7800 kg m^{-3}) falls through glycerol (ρ = 1260 kg m^{-3}, η = 1.4 Pa s). Find its terminal velocity.

$V = \frac{4}{3}\pi(1.0 \times 10^{-3})^3 = 4.19 \times 10^{-9}$ m^3.

$W = 7800 \times 4.19 \times 10^{-9} \times 9.81 = 3.21 \times 10^{-4}$ N. $U = 1260 \times 4.19 \times 10^{-9} \times 9.81 = 5.18 \times 10^{-5}$ N.

$F = W - U = 2.69 \times 10^{-4}$ N.

$v = F \div 6\pi\eta r = 2.69 \times 10^{-4} \div (6\pi \times 1.4 \times 1.0 \times 10^{-3}) =$ **1.0×10^{-2} m s^{-1}** (1.0 cm s^{-1}).

Low speed, small sphere: Stokes' law is reasonable here.

WORKED EXAMPLE: DOES STOKES' LAW APPLY?

A larger ball's measured speed is 1.7 m s^{-1}, but Stokes' law with $W - U$ predicts 0.050 m s^{-1}. The two don't agree, so Stokes' law **does not apply**: at this speed the flow is turbulent (WPH11 Jun 2024 Q14).

MARK-SCHEME LANGUAGE

- Conditions: "small, spherical object" or "spherical object at low speed"; "**laminar flow** (allow absence of turbulent flow)" (WPH11 Jan 2026 Q15, Oct 2024 Q19).
- Laminar flow: "no abrupt change in direction or speed of flow; layers don't mix; velocity at a point is constant".
- Temperature: "viscosity **decreases** as temperature increases" (or the converse).
- "Comparison of F with $W - U$ and conclusion consistent with the values" (WPH11 Jun 2022 Q19).
- Say "**viscous** drag": "drag" alone was not enough (WPH11 Oct 2020 Q15).

Often asked as: state a condition for Stokes' law; calculate F, η or v; "deduce whether Stokes' law applies" (5 marks); effect of temperature on viscosity and terminal velocity; effect of changing r or η; bubbles and balloons rising.

COMMON MISTAKE

Giving "terminal velocity", "constant temperature" or "known densities" as conditions for Stokes' law.
Giving "spherical" as a condition when the question already says the object is a sphere.
"The **object** must be laminar." It's the **flow** that is laminar.
Treating v in $6\pi\eta r v$ as a volume, or using the diameter as r.
Forgetting upthrust when finding the drag at terminal velocity.
Rounding the speed to 1 s.f. before using it.
Saying a higher viscosity increases the terminal velocity, or confusing viscosity with density.

BEYOND THE SPEC, BUT EXAMINED

- Unit of viscosity: Pa s, or kg m^{-1} s^{-1} in base units (WPH11 Jan 2023 Q12).
- Rearranged terminal-velocity formula: $v = 2r^2g(\rho_s - \rho_f) \div 9\eta$. It is not in the spec, and a misremembered version scores nothing, so it's safer to work from W = U + F each time (WPH11 Oct 2019 Q15).
- Scaling: $W \propto r^3$ and drag $\propto r$, so a smaller sphere is needed to keep the speed low in a less viscous fluid such as blood (WPH11 Jun 2022 Q19).

SPEC 26

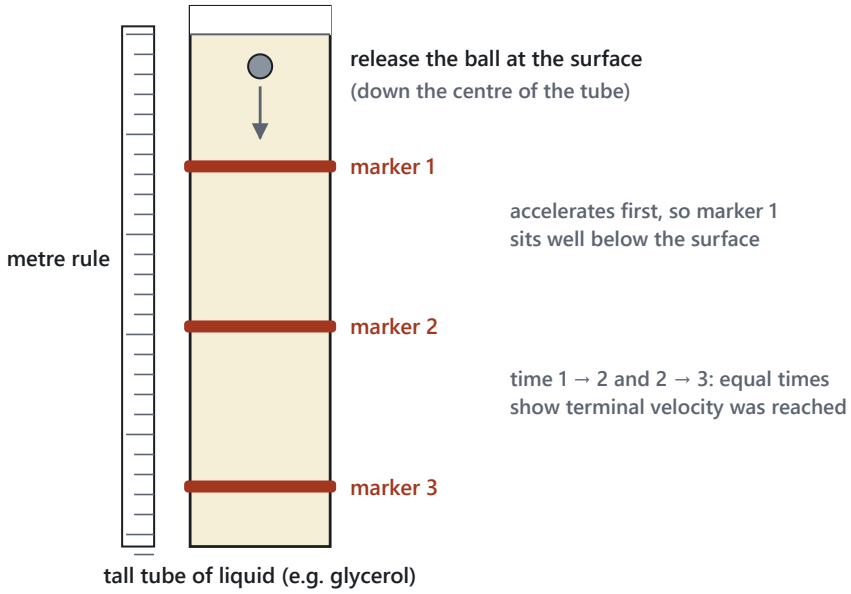
Core Practical 2: viscosity by the falling-ball method

WHAT THIS SPEC POINT ASKS

Use a falling-ball method to determine the viscosity of a liquid.

26 Principle

Time a ball bearing falling at **terminal velocity** through the liquid. Then use $W = U + 6\pi\eta rv$ to find η.



Falling-ball set-up. The first marker is well below the surface so the ball is already at terminal velocity; equal times between markers 1–2 and 2–3 confirm it.

CORE PRACTICAL 2: METHOD IN BRIEF

1. Measure the **mass** of a ball bearing (or several identical ones) on a balance, and its **diameter** with a micrometer or vernier calipers (several readings at different orientations; take a mean; halve for r).
2. Find the **density of the liquid**: mass of a measured volume (balance and measuring cylinder). Measure its **temperature**.
3. Fix rubber bands round a tall tube of the liquid as markers: the top one well **below the surface** (the ball accelerates at first), the others below it at measured equal spacings (e.g. 10–15 cm).
4. Release the ball **at the surface**, down the **centre** of the tube.
5. Time it between the markers with a stopwatch (or light gates / video for better timing). Equal times between successive pairs show it's at terminal velocity.
6. Repeat (at least 3 times) and use the mean time. v = distance ÷ mean time.

7. Calculate η from $W = U + 6\pi\eta rv$. For a better result, repeat with balls of different radii and plot v against r^2 (gradient = $2g(\rho_s - \rho_f) \div 9\eta$).

Safety: clamp the tall tube so it can't topple; wipe up spills (slippery); a magnet retrieves the balls.

26 Variables and errors

- **IV**: ball radius (if varied). **DV**: terminal velocity. **Control**: the liquid, its **temperature**, the ball material.
- **Ball not at terminal velocity** before the first marker → time too short, v too high, η too **low**. Fix: move the markers lower; check equal times.
- **Parallax** reading the ball passing a marker: eye level with the band; reaction time with a stopwatch → random error; use a longer distance or a video.
- **Wall effects** in a narrow tube slow the ball: use a wide tube and drop down the centre.
- A larger, faster ball gives a **shorter time**, so a bigger percentage uncertainty in the timing (WPH11 Oct 2019 Q15).

WORKED EXAMPLE: PROCESSING RESULTS

Ball radius r = 1.00 mm, ρ_s = 7800 kg m^{-3}; glycerol ρ_f = 1260 kg m^{-3}. Mean time to fall 0.150 m between markers = 15.0 s.

$v = 0.150 \div 15.0 = 0.0100$ m s^{-1}.

$\eta = 2r^2g(\rho_s - \rho_f) \div 9v = 2 \times (1.00 \times 10^{-3})^2 \times 9.81 \times (7800 - 1260) \div (9 \times 0.0100)$ = **1.4 Pa s**.

(This rearranged formula comes from $W = U + 6\pi\eta rv$ with W and U written as $\frac{4}{3}\pi r^3\rho g$.)

MARK-SCHEME LANGUAGE

- Method (5 marks, WPH11 Oct 2019 Q15): "two or more markers on the cylinder"; "top marker far enough below the surface for terminal velocity to be reached, or more than two markers and check the velocity is constant"; "measure the time to fall between markers (stopwatch) and the distance (metre rule)"; "repeat and average"; "v = distance ÷ mean time (or gradient of distance–time)".
- Why the first marker is low: "the sphere will be accelerating initially / needs a distance to reach terminal velocity" (WPH11 Oct 2020 Q15).
- Measurements: "mass of the ball bearing with a balance"; "diameter with (micrometer / vernier) calipers". Calipers measure the **diameter**, not the radius (WPH11 Jan 2023 Q12).

Often asked as: describe how to measure terminal velocity; why a third marker; which error makes η too high or too low; full viscosity calculations from given data; why a larger ball raises the percentage uncertainty.

COMMON MISTAKE

Dropping the ball from above the liquid surface.
Describing light gates or a camera when the question says stopwatch and metre rule.
Explaining how to calculate η when only the terminal velocity was asked.
"For accuracy" with no explanation of **what** is improved and **why**.

SPEC 27 Hooke's law and stiffness

WHAT THIS SPEC POINT ASKS

Use $\Delta F = k\Delta x$, where k is the stiffness of the object.

$$\Delta F = k\Delta x$$

ΔF = change in force (N), k = stiffness (spring constant) (N m^{-1}), Δx = extension or compression (m).

- **Hooke's law**: the extension is **proportional** to the force applied, **up to the limit of proportionality**. On a force–extension graph this is a straight line through the origin, and **k = gradient**.
- **Stiffness** k = force per unit extension. A large k means a stiff object (hard to stretch).
- k belongs to the **object**: a longer or thinner wire of the same material has a smaller k. (The Young modulus, 28, belongs to the **material**.)
- **Extension = new length − original length.** It also works for compression (Δx = shortening).

WORKED EXAMPLE: A SPRING

A spring extends from 12.0 cm to 16.0 cm when a 2.0 N weight hangs from it.
Δx = 4.0 cm = 0.040 m. k = $\Delta F \div \Delta x$ = 2.0 ÷ 0.040 = **50 N m^{-1}**.
With a 3.0 N weight: Δx = 3.0 ÷ 50 = 0.060 m, so the new length = 12.0 + 6.0 = **18.0 cm**.

27 Springs in series and parallel (beyond the spec, but examined)

	Series (end to end)	Parallel (side by side)
Force in each spring	the **full load**	load **shared** (half each, if identical)
Total extension	**sum** of the extensions	same as each spring's
Combined stiffness	$1/k = 1/k_1 + 1/k_2$	$k = k_1 + k_2$
Two identical springs of stiffness k	**k/2**	**2k**

WORKED EXAMPLE: SERIES AND PARALLEL

Two 50 N m^{-1} springs carry a 3.0 N load.
Series: k = 25 N m^{-1}, so Δx = 3.0 ÷ 25 = **0.12 m** (each spring carries 3.0 N and stretches 0.060 m).
Parallel: k = 100 N m^{-1}, so Δx = 3.0 ÷ 100 = **0.030 m** (each carries 1.5 N).

MARK-SCHEME LANGUAGE

- "Use of $\Delta F = k\Delta x$"; gradient "using a triangle over more than half the drawn line".
- Limit of proportionality: "the point beyond which force is **no longer proportional** to extension" (WPH11 Oct 2020 Q16). Don't also define the elastic limit.
- Series springs (4 marks, WPH11 Oct 2020 Q16): "both springs carry the full load; the extensions add; so for the same force the extension doubles; so the effective stiffness halves".
- "Stiffness" is the one-word answer for what k measures (WPH11 Oct 2025 Q16).

Often asked as: k from a graph or two lengths; new length under a load; springs in series or parallel (MCQs and explanations); four springs sharing a load (each takes a quarter); object v material (k v E); F against total length (gradient still k).

COMMON MISTAKE

Using the **new length** instead of the extension in $F = k\Delta x$.
Thinking the load is shared between springs in **series**.
Mixing up the limit of proportionality and the elastic limit.
Plotting length instead of extension: the line no longer passes through the origin (it shifts right) (WPH11 Oct 2020 Q16).
Leaving extensions in cm or mm.

BEYOND THE SPEC, BUT EXAMINED

- Springs in series and parallel (WPH11 Oct 2020 Q16, Jun 2022 Q7, Jun 2023 Q10).
- $k = EA \div L$ links stiffness to the Young modulus (WPH11 Jun 2021 Q17).
- If the gradient changes after loading past the elastic limit, the stiffness has changed (WPH11 Oct 2020 Q16).

SPEC 28

Stress, strain and the Young modulus

WHAT THIS SPEC POINT ASKS

Use tensile or compressive stress = force ÷ cross-sectional area, strain = extension ÷ original length, and Young modulus = stress ÷ strain.

$$\text{stress } \sigma = F \div A \qquad \text{strain } \varepsilon = \Delta x \div x \qquad \text{Young modulus } E = \sigma \div \varepsilon$$

Quantity	Meaning	Unit
stress σ	force per unit cross-sectional area	Pa (N m^{-2})
strain ε	extension ÷ original length	**none** (a ratio; sometimes as %)
Young modulus E	stress ÷ strain (in the straight-line region)	Pa

- **Tensile** stress stretches; **compressive** stress squashes. The same equations apply.
- The Young modulus describes the **stiffness of a material**. It doesn't depend on the sample's length or thickness. Metals: about 10^{10}–10^{11} Pa. Use it to identify a material from a data table.
- Combining the equations: $E = Fx \div A\Delta x$. Examiners advise working out stress and strain separately first.
- Cross-sectional area from a diameter: $A = \pi(d/2)^2$, with d in **metres**.

WORKED EXAMPLE: COPPER WIRE

A copper wire 2.00 m long and 0.40 mm in diameter extends by 6.5 mm when a 5.0 kg mass hangs from it.
$A = \pi \times (0.20 \times 10^{-3})^2 = 1.26 \times 10^{-7}$ m^2. $F = 5.0 \times 9.81 = 49.1$ N.
$\sigma = 49.1 \div 1.26 \times 10^{-7} = 3.9 \times 10^{8}$ Pa. $\varepsilon = 6.5 \times 10^{-3} \div 2.00 = 3.25 \times 10^{-3}$.
$E = 3.9 \times 10^{8} \div 3.25 \times 10^{-3} =$ **1.2×10^{11} Pa** (120 GPa).

MARK-SCHEME LANGUAGE

- "Use of σ = F/A", "use of ε = Δx/x", "use of E = σ/ε": separate marks. "Calculates cross-sectional area" first.
- Defining the Young modulus: the equation alone scores nothing unless **every term is defined**; "a measure of stiffness" is not enough (WPH11 Jan 2020 Q18).
- E (and breaking stress) are properties of the **material**, independent of shape and size (WPH11 Jan 2020 Q18).
- One scheme's working writes the force as 1.34×10^6 N where the question gives 13.4×10^6 N (WPH11 Jun 2022 Q12); use the question's value.

Often asked as: stress, strain or E from data; extension from a given E; "decide whether the cable will break" (compare stress with breaking stress); identify the material from a table; scaling MCQs (double the diameter → a quarter of the stress).

COMMON MISTAKE

Powers of ten: mm → m, mm^2 → m^2 ($\times 10^{-6}$), MPa (10^6) and GPa (10^9) (e.g. 110 MPa used as 110 Pa).
Using the diameter as the radius, or forgetting to square it. Very common.
Using the **new length** instead of the extension for strain.
Giving strain a unit; leaving the unit off E.
Using stress ÷ strain at the breaking point instead of in the straight-line region.

BEYOND THE SPEC, BUT EXAMINED

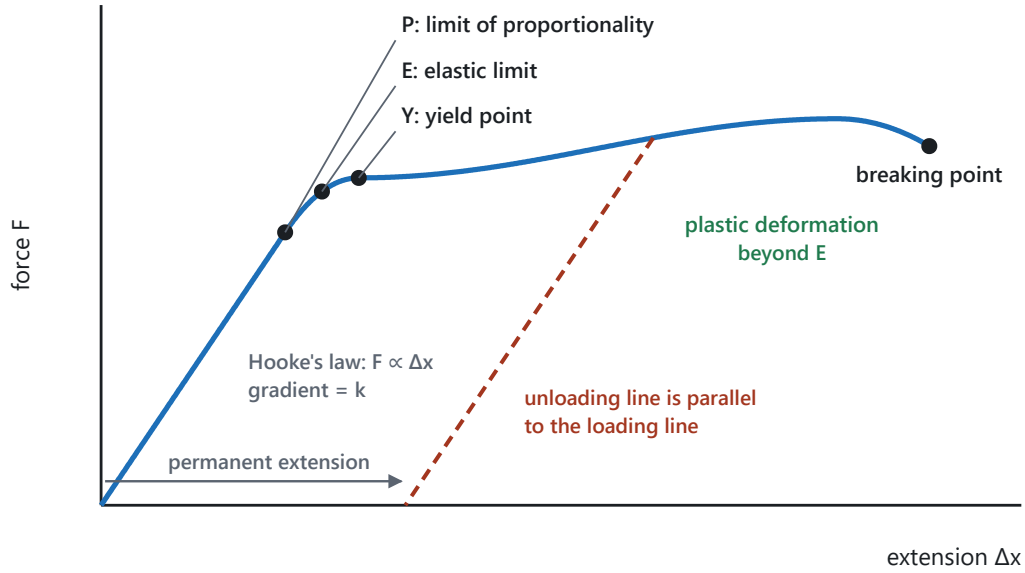
- Ratio questions: same material, different sizes → use E being the same (WPH11 Oct 2023 Q8).
- Stiffness of a sample from E: k = EA ÷ L. For the wire above, $k = 1.2 \times 10^{11} \times 1.26 \times 10^{-7} \div 2.00 = 7.5 \times 10^3$ N m^{-1} (WPH11 Jun 2021 Q17).

SPEC 29

Force–extension graphs: elastic and plastic deformation

WHAT THIS SPEC POINT ASKS

(a) Draw and interpret force–extension and force–compression graphs. (b) Use the terms limit of proportionality, elastic limit, yield point, elastic deformation and plastic deformation, and apply them to these graphs.



A metal wire. Loading beyond the elastic limit and then unloading gives a line parallel to the original one, leaving a permanent extension.

29 The key points (learn these definitions)

Term	Meaning
Limit of proportionality (P)	the point beyond which force is **no longer proportional** to extension (the graph stops being a straight line)
Elastic limit (E)	the force beyond which the object **will not return to its original length when the force is removed**
Yield point (Y)	the point where the material suddenly stretches a lot (a **large increase in extension**) for **little or no increase in force**; the extra extension is plastic
Elastic deformation	the object **returns to its original shape and size when the force is removed**
Plastic deformation	the object is **permanently deformed**: it does not return to its original shape when the force is removed

- P and E are close together, but they are **different** points: a material can still behave elastically a little beyond P.
- **Unloading**: below the elastic limit, the unloading line retraces the loading line back to the origin. Beyond it, the unloading line is (for a metal) **parallel to the original straight line** and meets the axis at a **permanent extension**.
- **Compression graphs** look similar, with "compression" on the x-axis (plotted as positive values).
- A graph that doesn't start at the origin (e.g. an initial load to straighten the wire): use **changes** in force and extension.

MARK-SCHEME LANGUAGE

- Elastic limit: "the force / stress beyond which the cable **does not return to its original length when the force is removed**" (WPH11 Jun 2022 Q12).
- Elastic deformation: "returns to its original size / shape / length **when the force is removed**" (WPH11 Jun 2025 Q15); "or the unloading line returns to the origin" (WPH11 Jan 2024 Q11).
- Yield point: "a **large** increase in strain for **little or no** increase in stress" **and** "the extension is plastic"; both parts are needed (WPH11 Jun 2023 Q20).
- Plastic: "remains deformed after the force is removed" (WPH11 Oct 2019 Q16).

Often asked as: define one of the five terms (1–2 marks); mark P or E on a graph; loading and unloading curves (rubber band); compression graphs (values in nm); MCQs ("beyond the elastic limit → permanent deformation").

COMMON MISTAKE

Leaving out "**when the force is removed**" from the elastic definitions. That's the mark.
Giving Hooke's law or the elastic limit when asked to define **elastic deformation**.
Treating the yield point as the elastic limit: the spec treats them as different points.
Forgetting to subtract the original length when plotting extension.

BEYOND THE SPEC, BUT EXAMINED

- The area **between** the loading and unloading curves = energy transferred to heat in the material (hysteresis), e.g. a rubber band (WPH11 Jan 2024 Q11). See 30 and 32.
- After plastic deformation, the gradient (stiffness) can change (WPH11 Oct 2020 Q16).

SPEC 30 Stress–strain graphs and breaking stress

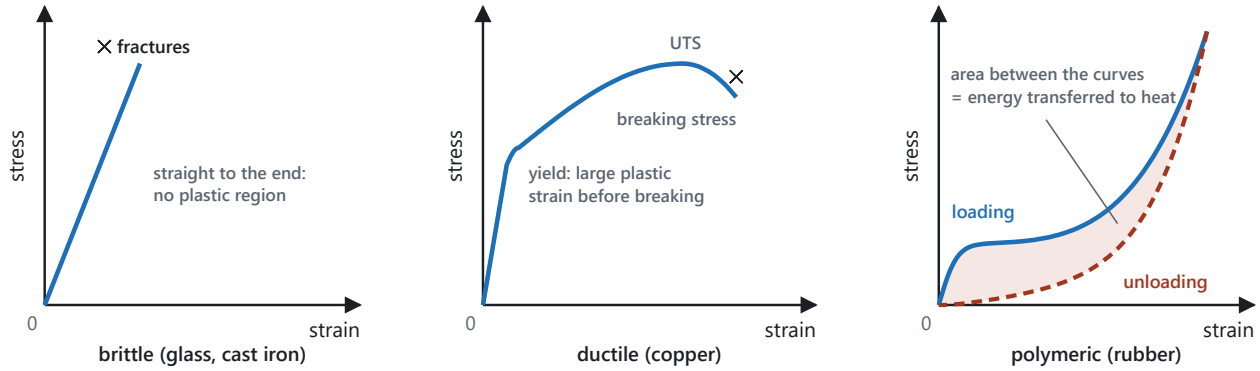
WHAT THIS SPEC POINT ASKS

Draw and interpret tensile or compressive stress–strain graphs, and understand the term breaking stress.

30 Reading a stress–strain graph

- **Gradient of the straight part = Young modulus** (stiffness of the material). Steeper = stiffer.
- **Limit of proportionality, elastic limit, yield point**: the same meanings as on a force–extension graph, but in terms of stress.
- **Breaking stress**: the stress at which the material **breaks** (fractures), the **end** of the graph.
- **Ultimate tensile stress (UTS)**: the **highest** stress on the graph. For a ductile metal, it is a little above the breaking stress, because the wire "necks" (thins) before it snaps.
- **Area under the graph = energy per unit volume** (32). A large area means the material absorbs a lot of energy before breaking (tough).

Because the axes are stress and strain, the graph doesn't depend on the sample's size: it describes the **material**.



Brittle: straight line to fracture. Ductile: yields, then a long plastic region before breaking. Rubber (polymeric): unloading curve below the loading curve; it returns to zero strain, and the shaded area is energy transferred to heat.

30 Three types of behaviour

Behaviour	Graph shape	Examples
Brittle	straight line (obeys Hooke's law) right up to fracture; **no plastic region**	glass, cast iron, ceramics, concrete
Ductile	straight, then yields and shows a **large plastic region** before breaking; can be drawn into wires	copper, mild steel
Polymeric (rubber)	not a straight line: soft at first, stiffening at large strain; very large **elastic** strain; unloading curve lies **below** the loading curve	rubber, many polymers

The spec doesn't ask you to define brittle, ductile, tough or hard, but questions use the words and expect you to read them off a graph: stiffness = initial gradient, strength = breaking stress (or UTS), toughness = area under the curve.

WORKED EXAMPLE: BREAKING FORCE

The copper wire in 28 ($A = 1.26 \times 10^{-7}$ m^2) has a breaking stress of 2.2×10^8 Pa. What is the largest mass it can hold?

$F = \sigma A = 2.2 \times 10^8 \times 1.26 \times 10^{-7} = 27.7$ N. $m = F \div g = 27.7 \div 9.81 =$ **2.8 kg**.

A thicker wire of the same copper has the **same** breaking stress but needs a **larger force** to break it.

MARK-SCHEME LANGUAGE

- Comparisons must be **comparative** and quote values from the graph: "higher breaking stress", "stiffer / greater gradient" ("withstands more" is ambiguous) (WPH11 Jan 2019 Q17).
- Same material, different diameter: "**breaking stress is unchanged**"; larger force needed to break (WPH11 Jun 2023 Q20).
- MCQ: breaking stress at the end of each line; yield stress at the top of the initial straight section (WPH11 Jan 2024 Q9).

Often asked as: compare two materials or a material in tension and compression from their graphs; read a breaking stress and calculate the breaking force; sketch a stiffer material's line; 6-mark "how does adding glue change stiffness, strength and toughness?" (WPH11 Jan 2020 Q14).

COMMON MISTAKE

Using the stress at the elastic limit instead of the breaking stress.
Confusing Young modulus (stiffness) with strength (breaking stress).
Describing without **quoting values** from the graph.
Answering from chemistry knowledge (glue, cement) instead of reading the graph.

BEYOND THE SPEC, BUT EXAMINED

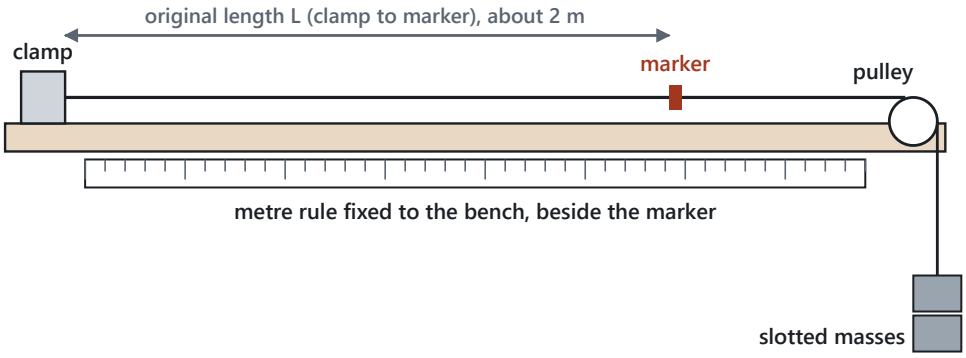
- UTS v breaking stress, and reading strength, stiffness and toughness from a graph (WPH11 Jan 2020 Q14).
- Pre-stressed concrete: steel rods stretched while the concrete sets keep the concrete **compressed**, so it can take larger tensile loads before cracking (WPH11 Oct 2019 Q16).
- Atomic picture: in the plastic region, layers of atoms slide over each other; at fracture, bonds break and atoms separate.

SPEC 31

Core Practical 3: the Young modulus of a material

WHAT THIS SPEC POINT ASKS

Determine the Young modulus of a material (usually a metal wire).



Horizontal-wire method. Load the wire in steps and read the marker's position against the fixed rule each time.

CORE PRACTICAL 3: METHOD IN BRIEF

1. Clamp one end of a long, thin wire (about 2 m) to the bench; pass the other end over a pulley and hang a mass hanger on it. Start with a small load to **straighten the wire** and take up the slack.
2. Fix a marker (tape or a small flag) on the wire near the pulley, above a metre rule fixed to the bench.
3. Measure the **original length** L from the clamp to the marker with a metre rule or tape.
4. Measure the **diameter** with a **micrometer** at several places along the wire, at two orientations at each place (right angles); take a mean. $A = \pi(d/2)^2$.
5. Add masses in equal steps (e.g. 100 g). Record the marker's position each time; **extension** = new reading − original reading.
6. Unload in steps and re-read, to check the wire returns (still elastic) and to give repeat readings.
7. Plot force (mg) against extension. In the straight part, gradient = $F/\Delta x$, and **E = gradient × L ÷ A**. (Or plot stress against strain: gradient = E.)

Safety: wear **eye protection** (a snapping wire can whip back); put a box of sand or a cushion under the masses; keep feet clear.
Vertical (Searle's) version: two wires hang from one support. The reference wire carries a constant load; a vernier scale between them measures the test wire's extension. This cancels the effects of temperature and of the support sagging.

31 Why a long, thin wire?

- For a given force, extension ∝ L ÷ A. A **long** wire gives a **bigger extension**, so the **percentage uncertainty** in the extension is smaller (the rule's absolute uncertainty stays the same).
- A **thin** wire needs a smaller force for a measurable extension, and reaches the stresses needed with ordinary slotted masses.

31 Uncertainties

- Diameter: use a **micrometer** (resolution 0.01 mm). Vernier calipers (0.1 mm or 0.02 mm) aren't good enough for a 0.4 mm wire: d is squared, so its percentage uncertainty doubles in A.
- Extension: read the marker at eye level against the rule (avoid **parallax**); a vernier scale gives better resolution.
- Use the gradient of the graph rather than one pair of readings.

WORKED EXAMPLE: E AND ITS UNCERTAINTY

L = 2.00 m, d = 0.40 mm, and the gradient of the force–extension line is 7.5×10^3 N m^{-1}.
A = π × $(0.20 \times 10^{-3})^2$ = 1.26×10^{-7} m^2. E = $7.5 \times 10^3 \times 2.00 \div 1.26 \times 10^{-7}$ = **1.2×10^{11} Pa**.
Percentage uncertainties for a single reading: d ± 0.01 mm → 2.5 %, so A → 2 × 2.5 = 5.0 %. L ± 2 mm → 0.1 %. Extension 6.5 mm ± 0.5 mm → 7.7 %.
Total ≈ 5.0 + 0.1 + 7.7 ≈ **13 %**. The extension is the biggest contributor, so a longer wire or a vernier scale would help most.

MARK-SCHEME LANGUAGE

- Diameter (3 marks): "micrometer (screw gauge) or digital calipers"; "at different places along the wire **and** different orientations"; "calculate a mean" (WPH11 Jun 2019 Q16, Jan 2024 Q17). Vernier calipers "do not have enough resolution" (WPH11 Jan 2021 Q16).
- Safety: "goggles, because the wire may break" (not helmets, gloves or shoes) (WPH11 Jan 2024 Q17).
- Full method (6-mark, WPH11 Jun 2024 Q16): original length clamp to marker; extension from the marker's movement (new − original); weight from mg; area from the diameter; stress and strain (or force–extension graph); E from stress ÷ strain or the gradient.
- Gradient: use more than half the drawn line; the origin may be used as a point if the line passes through it (WPH11 Jan 2023 Q14).
- Resolution: "a vernier scale has a better resolution, so a smaller percentage uncertainty" (WPH11 Jan 2024 Q17).

Often asked as: how to measure the diameter; safety; why a long thin wire (6-mark); full method (6-mark); E from a graph that doesn't start at the origin; resolution and percentage uncertainty.

COMMON MISTAKE

Measuring the diameter with a ruler or vernier, or only once.
Using "the centre of mass is in the middle" to check the wire is uniform: scored 0 (WPH11 Jan 2024 Q17).
Safety answers about anything except the snapping wire and falling masses.
Not noticing the graph doesn't start at the origin: use the change in force and the change in extension.
Using the words accuracy, precision, resolution and uncertainty as if they mean the same thing.

BEYOND THE SPEC, BUT EXAMINED

- Precise use of practical words: **resolution** is the smallest change an instrument can show; **uncertainty** is the range within which the true value is likely to lie; **precision** is how close repeated readings are; **accuracy** is how close a result is to the true value (WPH11 Jan 2024 Q17).
- Draw a tangent at the origin to find E when the straight-line part is short (WPH11 Jan 2019 Q17).

SPEC 32

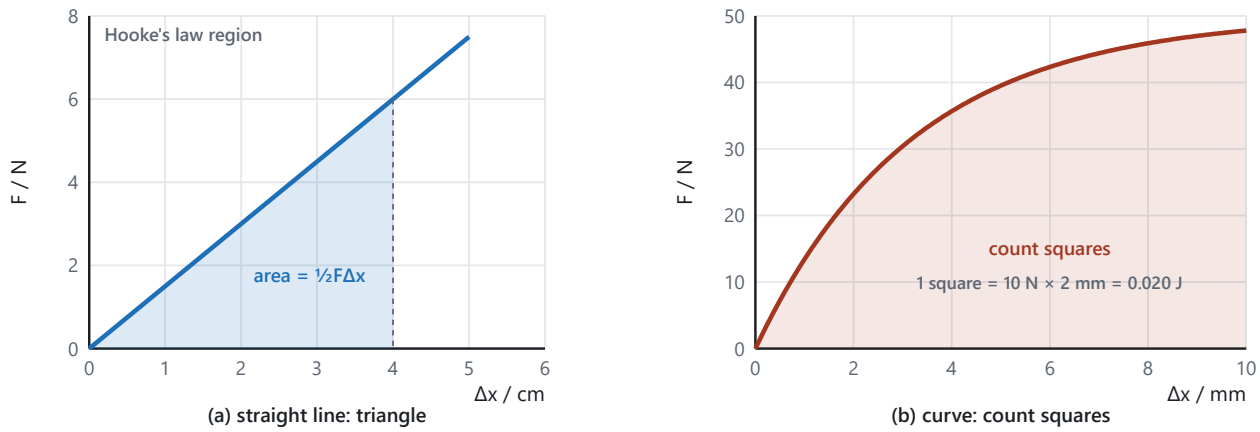
Elastic strain energy

WHAT THIS SPEC POINT ASKS

Calculate the elastic strain energy in a deformed sample, using $\Delta E_{el} = \frac{1}{2}F\Delta x$ and from the area under a force–extension graph, for linear and non-linear graphs.

32 Energy = area under the force–extension graph

The work done stretching a sample is stored as **elastic strain energy** (as long as the deformation is elastic). The work done = the **area under the force–extension graph**, for **any** shape of graph.



(a) Straight line (Hooke's law): the area is a triangle, ½FΔx. (b) Curve: count the squares and multiply by the energy one square represents.

For a straight line through the origin (Hooke's law):

$$\Delta E_{el} = \tfrac{1}{2}F\Delta x$$

Using $F = k\Delta x$ gives two other forms (not on the spec, but expected):

$$\Delta E_{el} = \tfrac{1}{2}k(\Delta x)^2 = F^2 \div 2k$$

The ½ is there because the force rises steadily from 0 to F: the **average** force is ½F.

WORKED EXAMPLE: A SPRING

The 50 N m^{-1} spring from 27 stretched 0.040 m by a 2.0 N force: $\Delta E_{el} = ½ \times 2.0 \times 0.040 =$ **0.040 J** (same as $½ \times 50 \times 0.040^2$).

WORKED EXAMPLE: CATAPULT (ENERGY CONSERVATION)

A catapult (k = 200 N m^{-1}) is pulled back 0.15 m and fires a 20 g stone. Find the launch speed, assuming all the elastic energy becomes KE.

$E_{el} = ½ \times 200 \times 0.15^2 = 2.25$ J. $½ \times 0.020 \times v^2 = 2.25 \rightarrow v = \sqrt{(2 \times 2.25 \div 0.020)} =$ **15 m s^{-1}**.

32 Non-linear graphs: estimate the area

- **Count squares**: count the squares under the curve (estimate the part-squares), then multiply by the energy of **one** square = (force per square) × (extension per square), in N and m.
- Or split the area into trapezia, triangles and rectangles.
- Graph (b) above: one large square = 10 N × 2 mm = 10 × 0.002 = 0.020 J. About 17 squares lie under the curve, so E ≈ 17 × 0.020 = **0.35 J**.
- Don't treat a curve as a straight line. In one question, a triangle approximation gave a 25 % reduction in energy where counting squares gave the expected 12–15 % (WPH11 Jan 2019 Q17).
- **Between two states** on a straight line (e.g. from 85 N to 140 N): work out ½FΔx at each and **subtract**. It is not ½ × ΔF × Δx (WPH11 Jun 2025 Q15).

32 Stress–strain graphs

- The area under a **stress–strain** graph = energy stored **per unit volume** (J m^{-3}). In the linear region it's ½σε.
- Multiply by the sample's volume (A × L) to get the energy. For the copper wire in 28: $½ \times 3.9 \times 10^8 \times 3.25 \times 10^{-3} = 6.3 \times 10^5$ J m^{-3}; $\times (1.26 \times 10^{-7} \times 2.00)$ m^3 = **0.16 J**, the same as $½F\Delta x = ½ \times 49.1 \times 6.5 \times 10^{-3}$.

32 Loading and unloading (hysteresis)

- If the unloading curve lies below the loading curve (rubber, or a metal past its elastic limit), not all the work done is given back. The **area between the curves** is the energy transferred to **heat** in the material.
- For plastic deformation, the work done goes into permanently rearranging the material, not into elastic strain energy.

MARK-SCHEME LANGUAGE

- "Use of $\Delta E_{el} = ½F\Delta x$"; "determines area under graph between …" with both values read (WPH11 Jun 2025 Q15).
- Area under stress–strain: "the elastic strain energy stored **per unit volume**"; then × volume (WPH11 Jan 2021 Q16, Jun 2026 Q19). Allow "elastic potential energy" for elastic strain energy.
- Deriving $F^2 \div 2k$ (3 marks): "E_{el} = area under the graph; area = ½FΔx and Δx = F/k; so $E_{el} = F^2/2k$" (WPH11 Oct 2025 Q15).
- Energy dissipated in a rubber band = area between the curves, e.g. a range 0.0425–0.053 J (WPH11 Jan 2024 Q11).

Often asked as: energy from a straight-line graph; area under a curve by counting squares; percentage change in energy absorbed; energy per unit volume from stress–strain; $½kx^2 = ½mv^2$ for a launcher; derive $F^2/2k$; energy lost in a loading–unloading loop.

COMMON MISTAKE

Forgetting the ½ (scored zero in WPH11 Jan 2022 Q16).
Treating a curve as a straight line and using ½FΔx.
Not converting cm, mm or nm to metres.
Using ΣF = ma with suvat during a spring release: the force isn't constant, so use energy.
Saying the stress–strain area **equals** the energy (it's energy per unit volume: multiply by the volume).
Forgetting to turn a result into a percentage when asked.

BEYOND THE SPEC, BUT EXAMINED

- ½k(Δx)^2 and F^2 ÷ 2k (WPH11 Oct 2025 Q15, Jan 2022 Q16).
- Hysteresis: area between loading and unloading curves = energy transferred to heat (WPH11 Jan 2024 Q11).
- Elastic energy combined with GPE and KE in one conservation question (WPH11 Jan 2026 Q19).